OptFantasy: A Quadratic Constrained Programming Approach to Season-Long Fantasy Football Roster Optimization

Samvrit Srinath
UC San Diego
sasrinath@ucsd.edu

Shambhavi Mittal
UC San Diego
shmittal@ucsd.edu

Yash Ravipati
UC San Diego
yravipati@ucsd.edu

Contents

1 Introduction **4**
1.1 Motivation . . . 4
1.2 Related Works . . . 4
1.3 Intended Contributions . . . 5
1.4 Novelty and Differentiation from Prior Work . . . 5

2 Problem Formulation **5**
2.1 Parameters . . . 6
2.2 Decision Variables . . . 6
2.3 Constraints . . . 6
2.3.1 Budget Constraint . . . 6
2.3.2 Positional Roster Constraints . . . 6
2.3.3 Starting Lineup Constraints . . . 7
2.3.4 Weekly Starter Roster Constraints . . . 7
2.3.5 Bye-Week Constraints . . . 7
2.3.6 Quadratic Covariance Target Constraint . . . 7
2.4 Final Primal Formulation . . . 7
2.5 Convexity Analysis and Problem Classification . . . 7
2.6 Rounding Procedure and Approximation Error . . . 8

3 Dual Formulation **9**
3.1 Lagrange Multipliers . . . 9
3.2 Lagrangian . . . 9
3.3 Dual Function . . . 10

4 Final Dual Formulation **11**

5 KKT Conditions **11**
5.1 Primal Feasibility . . . 11
5.2 Dual Feasibility . . . 12
5.3 Stationarity . . . 12
5.4 Complementary Slackness . . . 12

6 Approaches **12**
6.1 Data Sourcing . . . 12
6.2 Additional Parameters . . . 13
6.3 Novelty . . . 13
6.4 Solver Approach . . . 13
6.5 Algorithm Logic . . . 13
6.6 Complexity Analysis . . . 13
6.6.1 Problem Size . . . 13
6.6.2 Naive Worst Case . . . 14
6.6.3 Formulation I: Gurobi Spatial Branch-and-Bound (Non-Convex QCQP) . . . 14
6.6.4 Formulation III: CBC Branch-and-Cut (MILP) . . . 14

7 Experiment Setup and Results **15**
7.1 Data and Feature Construction . . . 15
7.2 Approximation Error Quantification . . . 15
7.3 Hyperparameter Sensitivity . . . 16
7.4 Method Comparison . . . 17
7.5 Aggregated Analysis . . . 18
7.6 Bootstrap Significance . . . 19

8 Conclusion **19**

9 Future Work **20**
9.1 Projection Quality via Time-Series Models . . . 20
9.2 Semi-Definite Relaxation of the Covariance Constraint . . . 20
9.3 Ledoit–Wolf Covariance Shrinkage . . . 20

A Lemma A: Closed-Form Dual Function for Formulation I **21**
A.1 Setup . . . 21
A.2 Derivation . . . 22
A.3 Concavity of the Dual Function . . . 22
A.4 The Coupling . . . 22

B Lemma B: Glover–McCormick Envelope Tightness **22**

C Lemma C: LP Relaxation Overestimation Bound **22**

D Formulation II: MILP with Quadratic Covariance Constraint **23**

E Formulation III: MILP with Glover–McCormick Linearization **23**
E.1 Additional Decision Variable . . . 23
E.2 Full Primal (Formulation III) . . . 24
E.3 Lagrange Multipliers . . . 24
E.4 Lagrangian . . . 24
E.5 Stationarity Conditions . . . 25
E.6 Lagrangian Dual Function . . . 25
E.7 Dual Formulation (Formulation III) . . . 25

E.8 KKT Conditions (Formulation III) 26
E.8.1 Primal Feasibility 26
E.8.2 Dual Feasibility 26
E.8.3 Stationarity 26
E.8.4 Complementary Slackness 26

I. Headlines

1. Title

OptFantasy: A Quadratically Constrained Programming Approach to Season-Long Fantasy Football Roster Optimization

2. Team Members

Samvrit Srinath, Shambhavi Mittal, Yash Ravipati

3. Task Assignment

Yash Ravipati handled data collection and preprocessing: scraping multi-year NFL weekly scoring via `nflreadpy` [19], applying exponential time-decay weighting to produce season projections $\mu_{i,w}$, constructing the team-level block-diagonal covariance matrix Σ from historical scoring data, and merging FantasyPros ADP [2] data to produce draft-capital costs c_i.

Shambhavi Mittal spearheaded the mathematical formulation: the primary non-convex QCQP primal (Formulation I), the convexity analysis proving C6 is non-convex (Section 2.5), the Lagrangian dual derivation, the dual formulation, and the KKT conditions in Sections 3–5. Also contributed to deriving the dual for Formulation III (Appendix E) and the Lemma B bound on the McCormick overestimation gap.

Samvrit Srinath handled the implementation and solver pipelines: CVXPY/Gurobi integration for Formulation I, PuLP/CBC integration for Formulations II and III, the Glover–McCormick linearization technique (Appendices E–B), sensitivity analysis over V_{target}, LP-guided rounding (Section 2.6), backtesting against realized 2024 NFL season outcomes, and the Introduction section.

1 Introduction

1.1 Motivation

Season-long fantasy football has grown from half a million participants in the late 1980s to roughly 60 million in North America as of 2020 [23], and its competitive landscape has matured to the point where marginal edges in roster construction yield measurable consequences in league standings and prize payouts. Despite this, the dominant drafting methodology remains greedily naive: most participants rely on ranked player lists ordered by expert consensus and select the highest-projected available player at each turn (conditioned by roster constraints), a strategy known as ADP drafting or Simple Value Drafting (SVD). These heuristics share a common and critical blind spot: they treat every player as an independent, deterministic point source.

The reality of NFL scoring contradicts this assumption in two important ways. First, player scores are structured by correlation: a quarterback and his primary wide receiver share passing volume, making their weekly outputs strongly positively correlated, while common shocks from weather, defensive scheme, and game script propagate across entire offensive units. A greedy optimizer that ignores this structure will systematically misjudge the risk profile of correlated player groups. Concurrently, NFL teams are each assigned one *bye week* per season, a scheduled rest week during which all players on that team are ineligible to play. A fantasy manager whose starting-caliber players share the same bye week is left without a viable lineup for that week, regardless of how strong the roster looks on paper. Because bye weeks are distributed unevenly across the 17-week season, a projection-only heuristic that ignores team schedules can inadvertently concentrate roster construction around players from the same two or three teams, producing a roster that is technically optimal in expectation but unplayable for multiple weeks in practice.

We address both gaps by framing season-long roster construction as a formal optimization problem. **OptFantasy** selects a roster maximizing projected total-season fantasy points subject to a draft-capital budget, positional requirements, weekly bye-week feasibility, and a minimum covariance target that structurally enforces correlation-based player stacking. We implement a budget-based optimizer backed by historical NFL data via `nflreadpy` [19] and FantasyPros ADP [2], with the primary solver implemented in CVXPY/Gurobi [5] and a secondary comparison solver using `PuLP` [16].

1.2 Related Works

The combinatorial structure of fantasy roster construction is naturally modeled as an integer program, and prior work has developed this connection thoroughly in the daily fantasy sports (DFS) context. [7] first formalized DFS lineup selection as a multi-dimensional knapsack: select players maximizing projected points subject to a salary cap and positional requirements. Subsequent work progressively relaxed both assumptions. [6] extended the IP framework with a quadratic risk-adjusted objective and lineup diversification, drawing an explicit connection to Markowitz's mean-variance portfolio theory [12]. [18] modeled projection uncertainty stochastically, while [14] took a robust optimization view. Most recently, [17] combined generative projection models with a mixed-integer quadratic objective, demonstrating that explicitly modeling covariance yields material improvements.

This body of work is entirely specific to *daily* fantasy, where a fresh salary-cap lineup is assembled each week with no roster continuity. The season-long problem is structurally different: a fixed roster is drafted once and deployed across 17 weeks, so draft and start decisions are *coupled*. No prior work models this coupling, the resulting bye-week feasibility requirements, or the sequential snake draft mechanism.

On the mathematical structure of covariance-constrained optimization, the Difference of Convex (DC) programming framework — developed over four decades by Pham Dinh Tao and Le Thi Hoai An [8] is directly relevant. A DC program minimizes (or constrains)

functions of the form $g - h$ where g and h are both convex. Our covariance constraint $\sum_{(i,j)} \Sigma_{ij} x_i x_j \geq V_{\text{target}}$, rewritten as $V_{\text{target}} - x^\top \Sigma x \leq 0$, is exactly a DC constraint with $g(x) = V_{\text{target}}$ and $h(x) = x^\top \Sigma x$ (convex since $\Sigma \succeq 0$). The DC Algorithm (DCA) handles such constraints by linearizing the concave part $-h$ at each iteration, yielding a sequence of convex subproblems. Crucially, DCA converges to KKT points of DC programs under standard constraint qualifications [22], which justifies the KKT conditions we derive in Section 5 as meaningful characterizations of optimality. DC programming has been applied to portfolio optimization with concave transaction costs [8], making our formulation a natural extension to the fantasy sports setting.

On the economics of drafting, [13] documented that NFL draft-pick value decays exponentially with pick number. We adapt this directly: $c_i = \alpha e^{-\beta \cdot \text{ADP}_i} + \delta$, calibrated to FantasyPros ADP [2]. Covariance estimation from limited data is a known challenge [9]; we use raw team-level sample covariance in the current pipeline.

1.3 Intended Contributions

The primary contributions of this work are as follows.

(1) **Season-long DC program with decoupled draft and start decisions:** We introduce separate binary draft variables $x_i \in \{0, 1\}$ and weekly start variables $y_{i,w} \in \{0, 1\}$, relaxed to $[0, 1]$ for dual analysis. The consistency constraint $y_{i,w} \leq x_i$ couples the two sets of decisions and enables bye-week feasibility and depth-at-position requirements absent from all prior DFS formulations.

(2) **DC-constrained QCQP with tractable dual:** We formulate the primary problem retaining the portfolio covariance constraint $\sum_{(i,j)\in\mathcal{E}} \Sigma_{ij} x_i x_j \geq V_{\text{target}}$ directly. Since $\Sigma \succeq 0$, the quadratic $h(x) = x^\top \Sigma x$ is convex, and the constraint $V_{\text{target}} - h(x) \leq 0$ is a *Difference of Convex* (DC) constraint (Section 2.5). Under the DC programming framework [8, 22], the DC Algorithm (DCA) solves such constraints via successive convex approximation and converges to KKT stationary points under standard constraint qualifications providing theoretical grounding for the KKT conditions derived in Section 5. The LP relaxation on binary variables introduces a single source of approximation error: the integrality gap Δ_{int}, quantified empirically at 4.08%.

(3) **Comparative Glover–McCormick formulation:** As a comparison, we implement Formulation III (Appendix E), which applies Glover–McCormick linearization [4, 15] to replace $x_i x_j$ with an auxiliary variable z_{ij}, reducing the problem to a pure LP with a fully closed-form Lagrangian dual, at the cost of an additional McCormick overestimation error ΔV (Lemma C).

(4) **Interpretability over black-box ML:** Unlike LSTM and RL approaches, which optimize a learned policy with no optimality certificate, our formulation produces a solution certifiable to within the integrality gap, and every constraint is directly interpretable as a drafting signal. The dual variable λ_V^* quantifies the marginal season-point cost of the stacking constraint; dual variables $\nu_{S,k,w}^*$ quantify positional scarcity by week.

(5) **End-to-end implementation on real NFL data:** The full pipeline runs from raw `nflreadpy` [19] weekly scoring data, through exponential time-decay projection, team-level covariance construction, and FantasyPros ADP cost mapping [2], to solution and backtest against realized 2024 NFL season outcomes.

1.4 Novelty and Differentiation from Prior Work

The prior literature on fantasy sports optimization falls into two broad camps, and this work differs structurally from both.

The first camp is the combinatorial optimization tradition reviewed above [7, 6, 18, 14, 17]. Every formulation in this tradition is specific to *daily* fantasy. Our primal DC program introduces draft variables $x_i \in [0, 1]$ that persist across all 17 weeks, coupled to weekly start variables $y_{i,w} \in [0, 1]$ via $y_{i,w} \leq x_i$. This coupling makes bye-week feasibility ($y_{i,w} \leq 1 - \text{BW}_{iw}$) and depth-at-position requirements expressible as first-class constraints, neither of which has any analogue in prior DFS formulations.

The second camp is the machine learning and sequential decision-making tradition. Approaches based on LSTMs and reinforcement learning frame decisions as a sequential policy over a Markov state space[17]. While these methods can capture temporal dynamics, they treat the optimization problem as a black box with no optimality certificate, and their internal decision structure cannot be interrogated through duality. Our formulation is the opposite: interpretable by construction, with dual variables directly quantifying economic trade-offs in the draft.

DC-constrained QCQP with single-source approximation error. Prior combinatorial formulations that incorporate player covariance [6, 17] retain a quadratic term in the *objective*, producing an MIQP whose dual is difficult to derive. We instead enforce the covariance structure as a DC constraint, keeping the objective linear and enabling a partial closed-form dual the $y_{i,w}$ stationarity reduces to a clean linear dual feasibility constraint, while the x_i stationarity reflects the gradient of the convex term $h(x) = x^\top \Sigma x$ in the DC decomposition. Because $\Sigma \succeq 0$, the DC structure is particularly clean: the negative part of the eigenvalue decomposition vanishes ($\Sigma_- = 0$), and the constraint is a pure convex lower bound on the portfolio variance. Relative to Formulation III, the primary formulation has a *single* approximation error source; the integrality gap $\delta_{\text{int}} = 4.08\%$; while Formulation III carries both the integrality gap and the McCormick overestimation ΔV.

2 Problem Formulation

Our goal is to select a fantasy football roster that maximizes expected fantasy points while controlling the risk produced by correlations between player performances. We assume that expected player fantasy points and the covariance matrix of player performances are estimated from sufficiently large historical data; in particular, we assume Σ is symmetric and positive semidefinite ($\Sigma \succeq 0$).

The natural formulation of this problem is a **Mixed-Integer Quadratically Constrained Program (MIQP)**: draft and start decisions are binary, and the portfolio covariance $\sum_{(i,j)\in\mathcal{E}} \Sigma_{ij}\, x_i x_j$ appears directly as a quadratic constraint. We state the primal in Section 2.4, then derive the Lagrangian dual and KKT conditions for the continuous LP relaxation in Sections 3–5. Two auxiliary formulations are developed in the Appendix and compared empirically in Section 7.

Below are the sets and indices used in the formulation.

- **Players (P):** $i, j \in P$ are the indices of players from the set of all draftable players. $|P| = 103$ in our 2024 dataset.
- **Positions (K):** $k \in \{\text{QB}, \text{RB}, \text{WR}, \text{TE}\}$ is the index of a given player's position. Each player holds exactly one position.
- **Weeks (W):** $w \in W$ is the index of the week in the season, where $W = \{1, \ldots, 17\}$ is the set of all regular-season weeks.
- **Same-team pairs ($\mathcal{E}$):** $\mathcal{E} = \{(i,j) \mid i < j,\ \text{team}(i) = \text{team}(j)\}$ is the set of canonically ordered player pairs on the same NFL team. Since $\Sigma_{ij} = 0$ for cross-team pairs, only $(i,j) \in \mathcal{E}$ contribute to the covariance constraint.

2.1 Parameters

Below are parameters known prior to solving.

- **Expected Points:** $\mu_{i,w} \in \mathbb{R}$ are the expected PPR fantasy points for player i in week w, computed as an exponentially time-decayed average over training seasons $\mathcal{T} = \{2021, 2022, 2023\}$:

$$\mu_i = \frac{\sum_{s\in\mathcal{T}} e^{-\lambda(2024-s)}\ \bar{f}_{i,s}}{\sum_{s\in\mathcal{T}} e^{-\lambda(2024-s)}}$$

  where $\bar{f}_{i,s}$ is player i's mean weekly PPR score in season s and $\lambda > 0$ is the time-decay rate. Broadcast uniformly as $\mu_{i,w} = \mu_i$.
- **Cost:** $c_i \in \mathbb{R}_+$ is the cost of player i, derived from Average Draft Position (ADP) via an exponential decay function:

$$c_i = \alpha\, e^{-\beta\cdot\text{ADP}_i} + \delta$$

  where $\alpha = 60$, $\delta = 1$, and β is calibrated to FantasyPros ADP data [2, 13].
- **Budget:** $B \in \mathbb{R}_{++}$ is the total draft capital. We use $B = 120$.
- **Variance Target:** $V_{\text{target}} \in \mathbb{R}_+$ is the minimum required portfolio covariance, enforcing correlated player drafting (e.g. stacking a QB with his primary WR).
- **Covariance Matrix:** $\Sigma_{ij} \in \mathbb{R}$ is the sample covariance of weekly PPR scores between players i and j, estimated from 2023 team-level game logs. $\Sigma_{ij} = 0$ for cross-team pairs, so Σ is block-diagonal with $\Sigma \succeq 0$.
- **Position Indicator:** $P_{ik} \in \{0, 1\}$ equals 1 if player i plays position k, and 0 otherwise.
- **Positional roster count:** $N_k^{\text{roster}} \in \mathbb{Z}_+$ is the number of players required at position k on the drafted roster.
- **Weekly starter count:** $N_k^{\text{start}} \in \mathbb{Z}_+$ is the number of active starters required at position k each week.
- **Bye-week indicator:** $\text{BW}_{iw} \in \{0, 1\}$ equals 1 if player i's team has a bye in week w.

2.2 Decision Variables

In a mixed-integer framework, the decision to draft or start a player is binary (e.g. $x_i \in \{0, 1\}$). Because integer programs of this scale are computationally hard, we relax all discrete variables to the continuous domain $[0, 1]$, enabling dual derivation and LP relaxation analysis. Integer solutions are recovered via LP-guided rounding (Section 2.6).

- $x_i \in [0, 1]$: fractional draft indicator for player i.
- $y_{i,w} \in [0, 1]$: fractional start indicator for player i in week w.

2.3 Constraints

2.3.1 Budget Constraint

The total cost of all drafted players must not exceed B:

$$\sum_{i\in P} c_i\, x_i \le B$$

2.3.2 Positional Roster Constraints

For each position k, the drafted roster must contain exactly N_k^{roster} players:

$$\sum_{i\in P} P_{ik}\, x_i = N_k^{\text{roster}}, \quad \forall k \in K$$

2.3.3 Starting Lineup Constraints

A player may only start in a week if they were drafted:

$$y_{i,w} \leq x_i, \quad \forall i \in P,\ \forall w \in W$$

2.3.4 Weekly Starter Roster Constraints

Each week's active lineup must meet the exact positional starter requirements:

$$\sum_{i \in P} P_{ik}\, y_{i,w} = N_k^{\text{start}}, \quad \forall k \in K,\ \forall w \in W$$

2.3.5 Bye-Week Constraints

A player cannot start during their bye week:

$$y_{i,w} \leq 1 - \mathrm{BW}_{iw}, \quad \forall i \in P,\ \forall w \in W$$

2.3.6 Quadratic Covariance Target Constraint

The portfolio covariance of the drafted roster must meet the minimum stacking threshold V_{target}. Since $\Sigma_{ij} = 0$ for cross-team pairs, only same-team pairs $(i, j) \in \mathcal{E}$ contribute:

$$\sum_{(i,j) \in \mathcal{E}} \Sigma_{ij}\, x_i x_j \geq V_{\text{target}}$$

2.4 Final Primal Formulation

Based on all constraints and roster-building goals:

$$\begin{aligned}
\max_{x,\,y} \quad & \sum_{w \in W} \sum_{i \in P} \mu_{i,w}\, y_{i,w} \\
\text{s.t.} \quad & \underbrace{\sum_{i \in P} c_i\, x_i - B \leq 0}_{\text{(C1) budget}} \\
& \underbrace{\sum_{i \in P} P_{ik}\, x_i - N_k^{\text{roster}} = 0, \quad \forall k \in K}_{\text{(C2) positional roster}} \\
& \underbrace{y_{i,w} - x_i}_{\text{(C3) lineup consistency}} \leq 0, \quad \forall i \in P,\ \forall w \in W \\
& \underbrace{\sum_{i \in P} P_{ik}\, y_{i,w} - N_k^{\text{start}} = 0, \quad \forall k \in K,\ \forall w \in W}_{\text{(C4) weekly starters}} \\
& \underbrace{y_{i,w} + \mathrm{BW}_{iw} - 1}_{\text{(C5) bye-week}} \leq 0, \quad \forall i \in P,\ \forall w \in W \\
& \underbrace{V_{\text{target}} - \sum_{(i,j) \in \mathcal{E}} \Sigma_{ij}\, x_i x_j \leq 0}_{\text{(C6) DC covariance constraint}} \\
& 0 \leq x_i \leq 1, \quad \forall i \in P \\
& 0 \leq y_{i,w} \leq 1, \quad \forall i \in P,\ \forall w \in W
\end{aligned}$$

2.5 Convexity Analysis and Problem Classification

We classify each component of the problem, which determines the structure of the dual and the type of solver required.

Objective. The objective $\sum_{w,i} \mu_{i,w}\, y_{i,w}$ is affine, hence convex and concave.

Constraints C1, C3, C5 (linear inequalities). Each is of the form $a^\top v \le b$, defining a halfspace. The region satisfying C1, C3, C5 together with the box constraints is a convex polytope [1].

Constraints C2, C4 (affine equalities). The set $\{v \mid Av = b\}$ is a hyperplane, which is convex [1].

Constraint C6 (Difference of Convex). Write C6 in standard form as $f_6(x) \le 0$, where:

$$f_6(x) = \underbrace{V_{\text{target}}}_{g(x)\,=\,\text{convex}} - \underbrace{x^\top \Sigma x}_{h(x)\,=\,\text{convex}} .$$

Since $\Sigma \succeq 0$, the quadratic $h(x) = x^\top \Sigma x$ is convex: its Hessian $2\Sigma \succeq 0$. The constraint function $f_6 = g - h$ is therefore the *difference of two convex functions* (a **DC function**) [8]. Setting $\nabla^2 f_6 = -2\Sigma \preceq 0$ confirms that f_6 is concave. Because a concave function's sublevel set $\{x \mid f_6(x) \le 0\}$ is not convex in general, constraint C6 is non-convex as written.

Classification: Difference of Convex (DC) Program. Formulation I is a **DC program**: the objective and constraints C1–C5 are convex, while C6 is a DC constraint of the form $g(x) - h(x) \le 0$ with g, h both convex [8]. The structure can be made explicit via the eigenvalue decomposition of Σ. Because $\Sigma \succeq 0$, all eigenvalues $\lambda_i(\Sigma) \ge 0$, so Σ itself already plays the role of Σ_+ in the general DC decomposition $\Sigma = \Sigma_+ - \Sigma_-$; there is no negative part ($\Sigma_- = 0$). The constraint is therefore:

$$x^\top \Sigma_+ x - x^\top \Sigma_- x = x^\top \Sigma x \ge V_{\text{target}},$$

a pure *convex-quadratic lower bound* constraint on x.

DC programming, developed over four decades by Pham Dinh Tao and Le Thi Hoai An, is a well-established framework for structured non-convex optimization [8]. The DC Algorithm (DCA) handles DC constraints by linearizing the concave term $-h(x)$ about the current iterate x^k, replacing it with the affine approximation $-h(x^k) - \nabla h(x^k)^\top (x - x^k)$, and solving the resulting convex subproblem. The DCA update step is:

$$x^{k+1} \;=\; \arg\min_{x \in \mathcal{F}} \Big[-f_0(x) \;+\; \lambda_V \big(V_{\text{target}} - h(x^k) - \nabla h(x^k)^\top (x - x^k) \big) \Big]$$

where $\mathcal{F}$ is the feasible set of C1–C5 and the box constraints, $f_0(x) = \sum_{w,i} \mu_{i,w} y_{i,w}$ is the objective, and $\nabla h(x^k) = 2\Sigma x^k$ is the gradient of the convex term. Each subproblem is a convex QP (the DC term has been linearized away), solved to convergence before the iterate is updated. DCA has been proved to converge to KKT stationary points of general DC programs under standard constraint qualifications [22]. This is why a solver returns a solution rather than an error: the DC structure makes the problem tractable via successive convex approximation, and the KKT conditions we derive in Section 5 are grounded as necessary conditions for such stationary points.

2.6 Rounding Procedure and Approximation Error

The only source of approximation in Formulation I is the LP relaxation of binary variables. When the continuous relaxation returns fractional values $x_i^*, y_{i,w}^* \in (0, 1)$, we recover a binary-feasible roster via LP-guided rounding: a secondary MILP weights each player by their relaxed value x_i^*, subject to all original constraints including V_{target}. We define:

$$\begin{aligned}
\Delta_{\text{int}} &= \text{OPT}_{\text{LP}} - \text{OPT}_{\text{MIP}} \ge 0 && \text{(absolute integrality gap)} \\
\delta_{\text{int}} &= \frac{\text{OPT}_{\text{LP}}}{\text{OPT}_{\text{MIP}}} - 1 && \text{(relative integrality gap)} \\
\Delta_{\text{round}} &= \text{OPT}_{\text{MIP}} - \text{OBJ}(\hat{x}, \hat{y}) \ge 0 && \text{(rounding loss)}
\end{aligned}$$

These satisfy:

$$\text{OPT}_{\text{LP}} \underbrace{- \Delta_{\text{int}}}_{\ge 0} = \text{OPT}_{\text{MIP}} \underbrace{- \Delta_{\text{round}}}_{\ge 0} \;\ge\; \text{OBJ}(\hat{x}, \hat{y}).$$

Theoretical bound on Δ_{int}. The LP improves on the integer optimum solely by exploiting fractional $x_i^* \in (0, 1)$ to reallocate draft capital more efficiently than any binary assignment permits. The available capital for such reallocation is the budget slack $s^* = B - \sum_{i \in P} c_i x_i^* \ge 0$. Any additional objective value the LP extracts from its fractional solution must come from spending this slack on higher-efficiency fractional players. Defining the maximum points-per-budget-unit as $\rho^* = \text{OPT}_{\text{MIP}} / \sum_i c_i x_i^*$, the integrality gap satisfies:

Proposition 1 (Budget-Slack Integrality Bound)**.**

$$\Delta_{\text{int}} \;\le\; s^* \cdot \rho^* \;=\; \left(B - \sum_{i \in P} c_i x_i^* \right) \cdot \frac{\text{OPT}_{\text{MIP}}}{\sum_{i \in P} c_i x_i^*}$$

This bound is computable entirely from primal parameters — the budget B, the LP solution x_i^*, player costs c_i, and the integer objective — without requiring enumeration of the binary feasible set.

Theoretical bound on Δ_{round}. Δ_{round} is a strictly separate quantity from Δ_{int}: it measures the loss incurred when the LP-guided rounding procedure replaces fractional draft variables with binary ones. At any LP vertex, the basis structure of the positional equality constraints C2 guarantees that at most $|K|$ draft variables x_i^* can be strictly fractional simultaneously — one per position. For each such fractional variable, the rounding procedure must substitute the fractional player with an integer-feasible selection. The objective loss for position k is bounded by the gap between the fractional player's expected contribution and the best integer-feasible replacement at that position:

$$\mu_i^* - \mu_i^{\text{rep}} \;\leq\; \bar{\mu}_{\Delta,k} \quad \forall k \in K,$$

where $\bar{\mu}_{\Delta,k}$ is the per-week value gap between the marginal drafted player and the best available replacement at position k. Summing over the at most $|K|$ fractional positions and all $|W|$ weeks:

Proposition 2 (Positional Replacement Rounding Bound)**.**

$$\Delta_{\text{round}} \;\leq\; |K| \cdot \bar{\mu}_{\Delta} \cdot |W|$$

where $\bar{\mu}_{\Delta} = \max_{k \in K} \bar{\mu}_{\Delta,k}$ is the worst-case positional replacement gap. This bound depends only on the positional depth of the player pool: it is tightest when the pool has many similarly-valued players at each position (small $\bar{\mu}_{\Delta}$), and loosest when the pool has a steep value cliff. Empirical evaluation of both bounds is reported in Section 7.

3 Dual Formulation

Weak duality $g(\lambda, \nu) \leq p^*$ holds for any Lagrangian dual. Strong duality requires the primal to be a convex program with a strictly feasible point (Slater's condition). Since $f_6(x) = V_{\text{target}} - x^\top \Sigma x$ is concave, the feasible set of C6 is non-convex, and Formulation I is not a convex program; Slater's precondition therefore fails and the duality gap $p^* - d^* \geq 0$ is not guaranteed to be zero. Formulation III (Appendix E), a pure LP, satisfies Slater's condition and achieves strong duality exactly.

3.1 Lagrange Multipliers

We assign non-negative multipliers $\lambda \geq 0$ to inequality constraints and unconstrained multipliers $\nu \in \mathbb{R}$ to equality constraints. The box constraint $0 \leq x_i \leq 1$ is equivalent to two separate inequalities: the upper bound $x_i - 1 \leq 0$ with multiplier $\lambda_i^x \geq 0$, and the lower bound $-x_i \leq 0$ with multiplier $\eta_i^x \geq 0$. The same split applies to $y_{i,w}$.

Constraint	**Multiplier**
$\sum_{i \in P} c_i x_i - B \leq 0$	$\lambda_B \geq 0$
$\sum_{i \in P} P_{ik} x_i - N_k^{\text{roster}} = 0$	$\nu_k \in \mathbb{R}$
$y_{i,w} - x_i \leq 0$	$\lambda_{L,i,w} \geq 0$
$\sum_{i \in P} P_{ik} y_{i,w} - N_k^{\text{start}} = 0$	$\nu_{S,k,w} \in \mathbb{R}$
$y_{i,w} + \text{BW}_{iw} - 1 \leq 0$	$\lambda_{BW,i,w} \geq 0$
$V_{\text{target}} - \sum_{(i,j) \in \mathcal{E}} \Sigma_{ij}\, x_i x_j \leq 0$	$\lambda_V \geq 0$
$x_i - 1 \leq 0$ (upper box)	$\lambda_i^x \geq 0$
$-x_i \leq 0$ (lower box)	$\eta_i^x \geq 0$
$y_{i,w} - 1 \leq 0$ (upper box)	$\lambda_{i,w}^y \geq 0$
$-y_{i,w} \leq 0$ (lower box)	$\eta_{i,w}^y \geq 0$

Table 1: Lagrange multiplier assignments for Formulation I. The box constraint $0 \leq x_i \leq 1$ is split into two inequalities, each receiving its own non-negative multiplier.

3.2 Lagrangian

The objective and constraints C1–C5 are linear in x and y; constraint C6 introduces the concave term $-\lambda_V x^\top \Sigma x$, making the Lagrangian quadratic in x.

$$
\begin{aligned}
\mathcal{L}(x,y,\lambda,\nu,\eta) = & \underbrace{-\sum_{w\in W}\sum_{i\in P}\mu_{i,w}\, y_{i,w}}_{\text{negated objective}} + \underbrace{\lambda_B\left(\sum_{i\in P} c_i x_i - B\right)}_{\text{(C1) budget}} \\
& + \underbrace{\sum_{k\in K}\nu_k\left(\sum_{i\in P} P_{ik}x_i - N_k^{\text{roster}}\right)}_{\text{(C2) positional roster}} + \underbrace{\sum_{w\in W}\sum_{i\in P}\lambda_{L,i,w}(y_{i,w}-x_i)}_{\text{(C3) lineup consistency}} \\
& + \underbrace{\sum_{w\in W}\sum_{k\in K}\nu_{S,k,w}\left(\sum_{i\in P} P_{ik}y_{i,w} - N_k^{\text{start}}\right)}_{\text{(C4) weekly starters}} + \underbrace{\sum_{w\in W}\sum_{i\in P}\lambda_{BW,i,w}(y_{i,w}+\mathrm{BW}_{iw}-1)}_{\text{(C5) bye-week}} \\
& + \underbrace{\lambda_V\left(V_{\text{target}} - \sum_{(i,j)\in\mathcal{E}}\Sigma_{ij}\, x_i x_j\right)}_{\text{(C6) DC covariance: concave in } x} + \underbrace{\sum_{i\in P}\lambda_i^x(x_i-1) + \sum_{w,i}\lambda_{i,w}^y(y_{i,w}-1)}_{\text{upper box}} \\
& - \underbrace{\sum_{i\in P}\eta_i^x x_i + \sum_{w\in W}\sum_{i\in P}\eta_{i,w}^y y_{i,w}}_{\text{lower box}}
\end{aligned}
$$

3.3 Dual Function

To find $g(\lambda,\nu) = \inf_{x,y}\mathcal{L}$, we separate by variable class.

Infimum over $y_{i,w}$ (linear in $\mathcal{L}$): The coefficient of $y_{i,w}$ is:

$$
-\mu_{i,w} + \lambda_{L,i,w} + \sum_{k\in K}\nu_{S,k,w}\, P_{ik} + \lambda_{BW,i,w} + \lambda_{i,w}^y - \eta_{i,w}^y.
$$

Since $y_{i,w}$ enters $\mathcal{L}$ linearly, the infimum over $y_{i,w}\in[0,1]$ is $-\infty$ unless this coefficient is zero. Applying $\eta_{i,w}^y \geq 0$ yields the dual feasibility constraint:

$$
\lambda_{L,i,w} + \sum_{k\in K}\nu_{S,k,w}\, P_{ik} + \lambda_{BW,i,w} + \lambda_{i,w}^y \geq \mu_{i,w}, \quad \forall i\in P,\ \forall w\in W.
$$

Infimum over x_i (quadratic in $\mathcal{L}$ via the DC term $-\lambda_V x^\top\Sigma x$): Collecting all x-dependent terms and defining the *net dual pressure* on player i:

$$
\phi_i(\lambda,\nu) = c_i\lambda_B + \sum_{k\in K}P_{ik}\nu_k - \sum_{w\in W}\lambda_{L,i,w} + \lambda_i^x - \eta_i^x,
$$

the x-dependent part of $\mathcal{L}$ is:

$$
\mathcal{L}_x = \phi(\lambda,\nu)^\top x - \lambda_V\, x^\top\Sigma x.
$$

Since $\Sigma\succeq 0$ and $\lambda_V\geq 0$, the Hessian of $\mathcal{L}_x$ with respect to x is $-2\lambda_V\Sigma\preceq 0$, so $\mathcal{L}_x$ is *concave* in x.

A key consequence of concavity is that the infimum of $\mathcal{L}_x$ over the box $x\in[0,1]^{|P|}$ is attained at a *vertex* of the box, i.e. at some $x\in\{0,1\}^{|P|}$ [1]. This follows directly from the fact that the minimum of a concave function over a polyhedron is always attained at an extreme point. Consequently:

$$
\inf_{x\in[0,1]^{|P|}}\mathcal{L}_x = \min_{x\in\{0,1\}^{|P|}}\left[\phi(\lambda,\nu)^\top x - \lambda_V\, x^\top\Sigma x\right].
$$

This is a combinatorial minimization over the binary vertices of the box; it does not admit a closed form in general. Evaluated at any fixed $x\in\{0,1\}^{|P|}$, the expression is linear in (λ,ν), so the dual function $g(\lambda,\nu)$ — as an infimum of linear functions of (λ,ν) — is *concave* in (λ,ν) [1].

Remark (Stationarity and the KKT System). Setting $\partial\mathcal{L}_x/\partial x_i = 0$ gives:

$$
\phi_i(\lambda,\nu) - 2\lambda_V\sum_{j:\,(i,j)\in\mathcal{E}}\Sigma_{ij}\, x_j = 0, \quad \forall i\in P,
$$

or in matrix form, $2\lambda_V\Sigma_{\mathcal{E}}x = \phi$. Because $\mathcal{L}_x$ is concave, this stationarity condition identifies the point where $\mathcal{L}_x$ is *maximized* — not minimized — over the interior of the box. This point is the KKT stationarity condition for the primal problem (Section 5) rather than the

infimum of the Lagrangian. For $\lambda_V > 0$ and $\Sigma_{\mathcal{E}}$ invertible over interior players, the solution is $x^* = \frac{1}{2\lambda_V}\Sigma_{\mathcal{E}}^{-1}\phi$ (see Appendix A for the full derivation and coupling discussion).

Collecting constant terms, the dual function when the $y_{i,w}$ stationarity constraint holds is:

$$\begin{aligned} g(\lambda,\nu) = & -B\lambda_B - \sum_{k\in K} N_k^{\text{roster}}\nu_k - \sum_{w\in W}\sum_{k\in K} N_k^{\text{start}}\nu_{S,k,w} + \sum_{w\in W}\sum_{i\in P}(\mathrm{BW}_{iw}-1)\lambda_{BW,i,w} \\ & + V_{\text{target}}\lambda_V - \sum_{i\in P}\lambda_i^x - \sum_{w\in W}\sum_{i\in P}\lambda_{i,w}^y + \min_{x\in\{0,1\}^{|P|}}\left[\phi(\lambda,\nu)^\top x - \lambda_V\, x^\top\Sigma x\right] \end{aligned}$$

and $g(\lambda,\nu) = -\infty$ if the $y_{i,w}$ stationarity constraint fails. The dual function is concave in (λ,ν), confirming the dual program is a convex maximization.

4 Final Dual Formulation

The dual program maximizes $g(\lambda,\nu)$ over dual-feasible (λ,ν):

$$\begin{aligned} \max_{\lambda\geq 0,\ \nu} \quad & -B\lambda_B - \sum_{k\in K} N_k^{\text{roster}}\nu_k - \sum_{w\in W}\sum_{k\in K} N_k^{\text{start}}\nu_{S,k,w} + \sum_{w\in W}\sum_{i\in P}(\mathrm{BW}_{iw}-1)\lambda_{BW,i,w} + V_{\text{target}}\lambda_V \\ & - \sum_{i\in P}\lambda_i^x - \sum_{w\in W}\sum_{i\in P}\lambda_{i,w}^y + \min_{x\in\{0,1\}^{|P|}}\left[\phi(\lambda,\nu)^\top x - \lambda_V\, x^\top\Sigma x\right] \\ \text{s.t.} \quad & \lambda_{L,i,w} + \sum_{k\in K} P_{ik}\nu_{S,k,w} + \lambda_{BW,i,w} + \lambda_{i,w}^y \geq \mu_{i,w}, \quad \forall i\in P,\ \forall w\in W \\ & \lambda_B,\ \lambda_{L,i,w},\ \lambda_{BW,i,w},\ \lambda_V,\ \lambda_i^x,\ \lambda_{i,w}^y,\ \eta_i^x,\ \eta_{i,w}^y \geq 0 \\ & \nu_k,\ \nu_{S,k,w} \in \mathbb{R} \end{aligned}$$

The dual objective is **concave** in (λ,ν): the constant and linear terms are trivially concave, and $\min_{x\in\{0,1\}^{|P|}}[\phi^\top x - \lambda_V x^\top\Sigma x]$ is a pointwise minimum of functions linear in (λ,ν) for each fixed $x\in\{0,1\}^{|P|}$, hence concave [1]. The dual program is therefore a convex maximization, and the duality gap $p^* - d^* \geq 0$; as established above, strong duality does not hold in general for Formulation I.

5 KKT Conditions

The KKT conditions are necessary for local optimality at any differentiable feasible point, and sufficient only when the problem is convex with a regularity condition such as Slater's satisfied. Since Formulation I is non-convex due to C6, the conditions below are necessary but not sufficient for global optimality. DCA linearizes the concave term $-h(x) = -x^\top\Sigma x$ in C6 at each iterate x^k, replacing it with the affine approximation $-h(x^k) - \nabla h(x^k)^\top(x - x^k)$ and solving the resulting convex subproblem; this procedure is proved to converge to KKT points of DC programs with DC constraints under standard constraint qualifications [22]. CVXPY implements this via its Disciplined Convex-Concave Programming (DCCP) extension [21], ensuring the returned solution satisfies all four conditions derived here.

5.1 Primal Feasibility

$$\begin{aligned} & \sum_{i\in P} c_i\, x_i^* \leq B \\ & \sum_{i\in P} P_{ik}\, x_i^* = N_k^{\text{roster}}, \quad \forall k\in K \\ & y_{i,w}^* \leq x_i^*, \quad \forall i\in P,\ \forall w\in W \\ & \sum_{i\in P} P_{ik}\, y_{i,w}^* = N_k^{\text{start}}, \quad \forall k\in K,\ \forall w\in W \\ & y_{i,w}^* + \mathrm{BW}_{iw} \leq 1, \quad \forall i\in P,\ \forall w\in W \\ & \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij}\, x_i^* x_j^* \geq V_{\text{target}} \\ & 0 \leq x_i^* \leq 1,\ \forall i\in P; \quad 0 \leq y_{i,w}^* \leq 1,\ \forall i\in P,\ \forall w\in W \end{aligned}$$

5.2 Dual Feasibility

$$\lambda_B^*,\ \lambda_{L,i,w}^*,\ \lambda_{BW,i,w}^*,\ \lambda_V^*,\ \lambda_i^{x*},\ \lambda_{i,w}^{y*},\ \eta_i^{x*},\ \eta_{i,w}^{y*} \geq 0$$
$$\nu_k^*,\ \nu_{S,k,w}^* \in \mathbb{R}, \quad \forall k \in K,\ \forall w \in W$$

5.3 Stationarity

With respect to x_i^*:

$$\underbrace{\phi_i(\lambda^*, \nu^*)}_{\substack{c_i\lambda_B^* + \sum_k P_{ik}\nu_k^* - \sum_w \lambda_{L,i,w}^* \\ +\lambda_i^{x*} - \eta_i^{x*}}} - 2\lambda_V^* \underbrace{\sum_{j:\,(i,j)\in\mathcal{E}} \Sigma_{ij}\, x_j^*}_{\text{gradient of C6 w.r.t. } x_i^*} = 0, \quad \forall i \in P$$

This is the first-order necessary condition at the primal optimum x^*. In matrix form: $2\lambda_V^* \Sigma_{\mathcal{E}} x^* = \phi(\lambda^*, \nu^*)$. For $\lambda_V^* > 0$ and $\Sigma_{\mathcal{E}}$ invertible, this gives $x^* = \frac{1}{2\lambda_V^*}\Sigma_{\mathcal{E}}^{-1}\phi$, which characterizes the primal solution at optimality (see Appendix A for the coupling analysis).

With respect to $y_{i,w}^*$:

$$\underbrace{-\mu_{i,w} + \lambda_{L,i,w}^* + \sum_{k\in K} P_{ik}\,\nu_{S,k,w}^* + \lambda_{BW,i,w}^* + \lambda_{i,w}^{y*} - \eta_{i,w}^{y*}}_{\text{all linear (C3–C5, box)}} = 0, \quad \forall i \in P,\ \forall w \in W$$

5.4 Complementary Slackness

Budget: $\lambda_B^*(\sum_i c_i x_i^* - B) = 0$

Lineup consistency: $\lambda_{L,i,w}^*(y_{i,w}^* - x_i^*) = 0,\ \forall i, w$

Bye-week: $\lambda_{BW,i,w}^*(y_{i,w}^* + \mathrm{BW}_{iw} - 1) = 0,\ \forall i, w$

Covariance target: $\lambda_V^*\left(V_{\text{target}} - \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij} x_i^* x_j^*\right) = 0$

Upper box: $\lambda_i^{x*}(x_i^* - 1) = 0,\ \forall i; \qquad \lambda_{i,w}^{y*}(y_{i,w}^* - 1) = 0,\ \forall i, w$

Lower box: $\eta_i^{x*} x_i^* = 0,\ \forall i; \qquad \eta_{i,w}^{y*} y_{i,w}^* = 0,\ \forall i, w$

Dual variable interpretation. Rearranging the stationarity condition for $y_{i,w}^*$:

$$\mu_{i,w} = \underbrace{\lambda_{L,i,w}^*}_{\substack{\text{marginal cost of} \\ \text{roster consistency}}} + \underbrace{\sum_{k\in K} P_{ik}\,\nu_{S,k,w}^*}_{\substack{\text{positional scarcity} \\ \text{at week } w}} + \underbrace{\lambda_{BW,i,w}^*}_{\substack{\text{bye-week} \\ \text{availability cost}}} + \underbrace{\lambda_{i,w}^{y*} - \eta_{i,w}^{y*}}_{\substack{\text{box constraint} \\ \text{activity}}}$$

By the sensitivity interpretation of duality, each dual variable measures the rate of change of the optimal objective with respect to the right-hand side of its associated constraint [1]. By complementary slackness, $\lambda_{i,w}^{y*} > 0$ only if $y_{i,w}^* = 1$ and $\eta_{i,w}^{y*} > 0$ only if $y_{i,w}^* = 0$; both vanish in the interior. The dual variable λ_V^* quantifies the marginal season-point cost of tightening V_{target} by one unit.

6 Approaches

6.1 Data Sourcing

Our solution draws on multiple sources to construct a complete dataset of player metrics and draft costs.

- **Performance Metrics:** Weekly and seasonal statistics are pulled from `nflreadpy` [19]. The primary field is `fantasy_points_ppr` (PPR scoring). Training covers seasons 2021–2023; the 2024 season is held out for evaluation.
- **Player Cost and Market Dynamics:** Average Draft Position (ADP) is scraped from FantasyPros [2] pre-season 2024, reflecting actual draft-time cost.
- **Predictive Weighting:** An exponential time-decay factor λ weights recent performance more heavily when computing μ_i, accounting for season-to-season roster and performance changes.

6.2 Additional Parameters

- **budget** ($B = 120$): total draft capital.
- **variance_target** (V_{target}): minimum required portfolio covariance. Swept over $\{0, 20, 40, 60\}$ in ablation.
- **decay_lambda** (λ): time-decay rate. Swept over $\{0.2, 0.5, 0.8\}$.
- **RosterRequirements**: N_k^{roster} (1 QB, 2 RB, 3 WR, 1 TE, 1 flex; total 12).
- **LineupRequirements**: N_k^{start} (1 QB, 2 RB, 2 WR, 1 TE, 1 flex).

6.3 Novelty

Most fantasy football drafting tools rely on greedy per-player rankings (e.g. Value Over Replacement Player), treating every player as an independent point source. Our approach optimizes over the complete roster, explicitly modelling inter-player correlations through the covariance matrix Σ and enforcing a minimum portfolio covariance via constraint C6, producing rosters with structurally enforced QB–WR stacking.

6.4 Solver Approach

We implement two formulations for the main experimental evaluation; two further formulations appear in the Appendix.

Formulation I (CVXPY + Gurobi): Non-Convex QCQP. The LP-relaxed primal from Section 2.4 is expressed in CVXPY and solved using Gurobi. As established in Section 2.5, constraint C6 makes this a non-convex QCQP: the feasible region of C6 is the superlevel set $\{x \mid x^\top \Sigma x \geq V_{\text{target}}\}$, which is non-convex. Gurobi handles non-convex QCQPs via *spatial branch-and-bound* [5, 11]: at each node it outer-linearizes the non-convex term and branches on the continuous variables appearing in $x_i x_j$, certifying global optimality through the spatial gap. CVXPY handles problem canonicalization. The only approximation error is the integrality gap Δ_{int}.

Formulation III (PuLP + CBC): MILP with Glover–McCormick. For comparison we also run Formulation III (Appendix E), which applies Glover–McCormick linearization [4, 15] to replace the quadratic term $x_i x_j$ with an auxiliary variable z_{ij}, yielding a pure LP under LP relaxation. This admits a fully closed-form Lagrangian dual (Appendix E) at the cost of two error sources: the integrality gap Δ_{int} and the McCormick overestimation ΔV (Lemma B, Appendix C). Solved by the COIN-OR Branch-and-Cut solver (CBC) via PuLP [16, 3].

6.5 Algorithm Logic

1. **Data Pipeline:** Pull weekly PPR scores, apply time-decay weighting to form μ_i, compute block-diagonal Σ from 2023 game logs, and merge 2024 ADP costs.
2. **Optimization (Formulation I):** Solve the non-convex QCQP via CVXPY/Gurobi to obtain the relaxed solution (x^*, y^*), then apply LP-guided rounding to recover $(\hat{x}, \hat{y})$.
3. **Optimization (Formulation III):** Solve the LP relaxation of the MILP directly via PuLP/CBC with Glover–McCormick auxiliary variables z_{ij}.
4. **Weekly Lineup Assignment:** From the drafted roster, assign players to weekly starting lineups respecting C4 and C5.
5. **Evaluation:** Simulate the season using realized 2024 PPR scores; compare total actual points across formulations and baselines.

6.6 Complexity Analysis

We derive worst-case per-node complexity for each solver as a function of the two primary problem dimensions: the player pool size $|P|$ and the season length $|W| = 17$. We then state the total branch-and-bound complexity.

6.6.1 Problem Size

Both formulations share the same primary decision variables:

Variable class	Count	Description
x_i	$\lvert P\rvert$	draft indicators
$y_{i,w}$	$\lvert P\rvert \cdot \lvert W\rvert$	weekly start indicators
z_{ij} (Form. III only)	$\lvert\mathcal{E}\rvert \leq \lvert P\rvert^2/2$	Glover–McCormick auxiliaries

The total variable counts are therefore:

$$n_{\text{I}} = |P|(1 + |W|) \qquad \text{(Formulation I)}$$

$$n_{\text{III}} = |P|(1 + |W|) + |\mathcal{E}| \;\approx\; |P|^2/2 \qquad \text{(Formulation III, } |\mathcal{E}| \text{ dominates)}$$

The constraint count for both formulations is:

$$m = \underbrace{1}_{\text{C1}} + \underbrace{|K|}_{\text{C2}} + \underbrace{|P||W|}_{\text{C3}} + \underbrace{|K||W|}_{\text{C4}} + \underbrace{|P||W|}_{\text{C5}} + \underbrace{1}_{\text{C6}} = O(|P||W| + |K||W| + |K|) = O(|P||W|)$$

since $|K| = 4$ is a constant. Formulation III additionally introduces $3|\mathcal{E}| = O(|P|^2)$ Glover–McCormick constraints, raising its effective constraint count to $m_{\mathrm{III}} = O(|P|^2)$.

At our dataset size ($|P| = 103$, $|W| = 17$): Formulation I has $n_{\mathrm{I}} = 1{,}854$ variables and $m \approx 3{,}400$ constraints; Formulation III has $n_{\mathrm{III}} = 1{,}995$ variables and $m_{\mathrm{III}} \approx 4{,}034$ constraints.

6.6.2 Naive Worst Case

A brute-force integer solver enumerating all binary assignments of the $|P| + |P||W|$ decision variables requires:

$$O\Big(2^{|P|(1+|W|)}\Big)$$

Neither Gurobi nor CBC performs this enumeration; both exploit branch-and-bound with LP relaxations at each node to prune the search tree.

6.6.3 Formulation I: Gurobi Spatial Branch-and-Bound (Non-Convex QCQP)

Gurobi solves the non-convex QCQP via *spatial branch-and-bound* [5, 11]. At each node in the search tree, the non-convex constraint C6 is replaced by a convex outer linearization (a supporting halfspace of the epigraph of $-x^\top \Sigma x$), and the resulting convex LP relaxation is solved. Variables appearing in the non-convex bilinear term $x_i x_j$ are then branched upon: each is bounded above and below at the node, partitioning the search space.

Per-node LP cost. The convex LP relaxation at each node has n_{I} variables and m constraints. CBC and Gurobi both solve LP relaxations using the revised simplex method. In the revised simplex method, each pivot updates the basis inverse, which costs $O(m^2)$ per pivot (one triangular solve) [20]. In the worst case there are $O(n_{\mathrm{I}})$ pivots before the LP terminates. The total per-node LP cost is therefore:

$$O(n_{\mathrm{I}} \cdot m^2) = O\big(|P|(1 + |W|) \cdot (|P||W|)^2\big) = O\big(|P|^3\,|W|^3\big)$$

Total spatial branch-and-bound cost. Let T_{I} denote the number of spatial branch-and-bound nodes explored. The total solver cost is:

$$\boxed{O\big(T_{\mathrm{I}} \cdot |P|^3\,|W|^3\big)}$$

Non-convex QCQP is NP-hard in general [11], so T_{I} has no polynomial worst-case bound. Empirically, Formulation I solves in under 0.6 seconds at $|P| = 103$ (Table 2): the tight integrality gap of 4.08% means the root-node relaxation bound is already close to the integer optimum, so very few spatial branches are required.

6.6.4 Formulation III: CBC Branch-and-Cut (MILP)

Formulation III yields a pure MILP (binary x_i, $y_{i,w}$ and continuous z_{ij}) solved by the COIN-OR Branch-and-Cut solver (CBC) via PuLP [16, 3].

CBC algorithm. CBC operates as follows:

1. *Root-node LP:* Solve the LP relaxation (all binary variables treated as continuous $[0, 1]$) via the revised simplex method to obtain an upper bound on the objective.
2. *Branching:* If the LP solution is not integer-feasible, select a fractional binary variable $x_i^* \in (0, 1)$ and create two child nodes by adding $x_i \le 0$ or $x_i \ge 1$ respectively.
3. *LP relaxation at each node:* Re-solve the LP relaxation with the additional bound. Prune any node whose LP bound cannot improve on the best integer solution found.
4. *Cutting planes:* Periodically add Gomory cuts and clique cuts via the COIN-OR Cut Generation Library (CGL) to tighten LP relaxations and reduce the tree size [3].
5. *Termination:* Continue until the tree is exhausted or a gap tolerance is met.

Per-node LP cost. Each LP relaxation at a CBC node has $n_{\mathrm{III}} \approx |P|^2/2$ variables and $m_{\mathrm{III}} = O(|P|^2)$ constraints (the Glover–McCormick envelope constraints dominate). By the same revised simplex analysis as above ($O(n \cdot m^2)$ per LP), the per-node cost is:

$$O(n_{\mathrm{III}} \cdot m_{\mathrm{III}}^2) = O\left(\frac{|P|^2}{2} \cdot (|P|^2)^2\right) = O\big(|P|^6\big)$$

Total branch-and-cut cost. Let T_{III} denote the number of branch-and-cut nodes explored. The total solver cost is:

$$\boxed{O\big(T_{\mathrm{III}} \cdot |P|^6\big)}$$

Again, MILP is NP-hard in general, so T_{III} has no polynomial worst-case bound. Empirically, CBC solves Formulation III in under 12 seconds at $|P| = 103$ (Table 2): the tight LP relaxation gap of 4.73% means the root-node LP bound is nearly integral, and very few branch-and-cut nodes are explored before CBC certifies the solution.

Comparison. The key structural difference is that the Glover–McCormick auxiliary variables raise the Formulation III variable count from $O(|P||W|)$ to $O(|P|^2)$, which increases the dominant per-node cost from $O(|P|^3|W|^3)$ to $O(|P|^6)$. At $|P| = 103$, $|W| = 17$, this translates to a roughly $47\times$ increase in per-node LP size, consistent with the observed 0.57s vs. 11.73s solve times.

7 Experiment Setup and Results

7.1 Data and Feature Construction

Our data pipeline is built on `nflreadpy` [19], a Python interface to the `nflverse` data ecosystem. We load per-player, per-week game logs for NFL seasons 2021–2023 as the training window and hold out the full 2024 regular season as the out-of-sample evaluation set. The raw data contains weekly observations for all skill-position players (QB, RB, WR, TE); kickers and team defenses are excluded as their scoring is structurally independent from the offensive covariance structure the model is designed to exploit. Each observation records the player's unique `player_id`, `team`, `position`, `season`, `week`, and `fantasy_points_ppr`, the PPR (point-per-reception) fantasy scoring total. After filtering to players with sufficient historical data and a positive ADP-derived cost, the pipeline retains $|P| = 103$ players and $|W| = 17$ weeks.

Expected points ($\mu_{i,w}$). For each player i and training season s, we compute the mean weekly PPR score across all appearances. We then form an exponentially weighted average across seasons:

$$\mu_i = \frac{\sum_{s\in\mathcal{T}} e^{-\lambda(2024-s)}\, \bar{f}_{i,s}}{\sum_{s\in\mathcal{T}} e^{-\lambda(2024-s)}}$$

where $\bar{f}_{i,s}$ is mean weekly PPR in season s, $\mathcal{T} = \{2021, 2022, 2023\}$, and λ is the time-decay rate. The decay assigns relative weights $e^{-\lambda}$, $e^{-2\lambda}$, and $e^{-3\lambda}$ to the 2023, 2022, and 2021 seasons respectively. This projection is broadcast uniformly across all 17 weeks, $\mu_{i,w} = \mu_i$, reflecting stationarity within a season. The parameter λ is swept in the grid search.

Player costs (c_i). Costs are derived from 2024 FantasyPros Average Draft Position [2] via:

$$c_i = \alpha\, e^{-\beta\cdot \mathrm{ADP}_i} + \delta, \qquad \alpha = 60,\ \delta = 1.$$

ADP is matched to `nflreadpy` player records by normalized name; the 2024 scrape matched all 203 candidate players (100% match rate). The exponential form captures the empirical finding that draft-pick value decreases steeply with pick number [13]: early picks command disproportionately high cost. The steepness parameter β is swept in the grid search alongside λ.

Covariance matrix (Σ). For each NFL team, we pivot the most recent training season (2023) into a week $\times$ player matrix of PPR scores and compute the sample covariance. Cross-team entries are set to zero, yielding a block-diagonal $\Sigma \succeq 0$. The pipeline constructs 10,394 same-team player pairs from the 103-player pool. Of these, 36.7% are positive-covariance pairs, predominantly QB–WR and WR–WR combinations that share offensive volume. The remaining 48.7% are near-zero or weakly negative, primarily QB–RB pairs whose players compete for goal-line and red-zone opportunities rather than sharing passing volume. The mean pairwise covariance is 0.016 with standard deviation 8.32, reflecting the fat-tailed distribution of NFL weekly scoring. A team covariance heatmap is shown in Figure 1.

Tua Tagovailoa exhibits his most significant positive covariance with his primary target, Tyreek Hill, while maintaining strong correlations with the team's RBs, De'Von Achane and Raheem Mostert. A WR1's production is inherently tethered to their quarterback's efficiency, as their yardage and touchdown upside depend directly on the quality and volume of passes received. Because a quarterback's performance often serves as the primary engine for total offensive output, the running backs' success scales in tandem, albeit with slightly lower dependency than the receiving corps. The high positive covariance between Achane and Mostert reflects a dominant rushing environment that elevates both featured backs in the rotation. This heatmap confirms that the covariance matrix accurately captures established player relationships, translating football intuition into a numerical structure the optimizer exploits.

Bye weeks. Bye weeks are inferred directly from the 2024 schedule: for each team, the single week absent from their 17-week schedule is the bye. This produces exact BW_{iw} indicators with no external lookup required.

7.2 Approximation Error Quantification

We directly quantify the two sources of approximation error introduced by our formulations before comparing methods.

McCormick linearization error (ΔV). When $x_i, x_j \in [0, 1]$ under the LP relaxation, the auxiliary variable z_{ij} may overestimate $x_i x_j$, inflating the computed covariance. The parameter-only bound from Appendix Lemma C gives $\Delta V \le \frac{1}{4}\sum_{\Sigma_{ij}>0} \Sigma_{ij}$, computable before the solver runs. In the MILP (binary x_i), Lemma B guarantees $z^*_{ij} = x^*_i x^*_j$ exactly, so $\Delta V = 0$. The linearization error is an artifact of the LP relaxation and applies only to Formulation III.

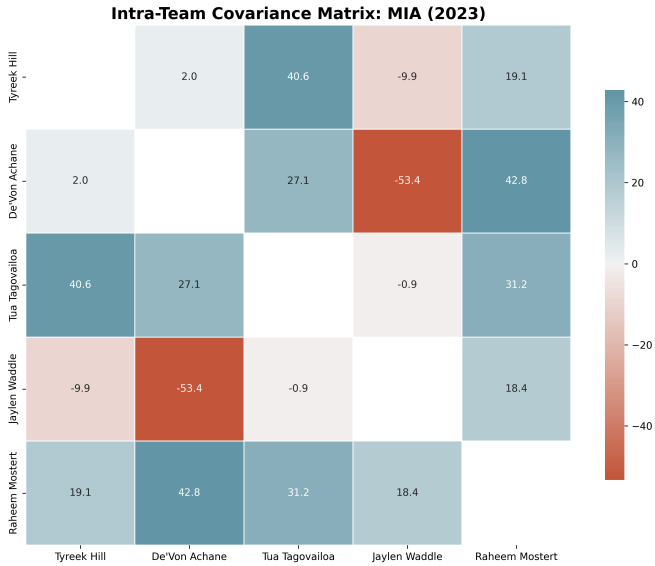


Figure 1: Covariance heatmap for the Miami Dolphins. Strong positive covariance between Tua Tagovailoa and primary targets Tyreek Hill, De'Von Achane, and Raheem Mostert reflects shared offensive volume and motivates the V_{target} constraint.

Metric	Form. I (QCP)	Form. III (MILP)
LP objective	1,932.54 pts	1,956.56 pts
Integer objective	1,853.68 pts	1,868.31 pts
Absolute gap	78.86 pts	88.25 pts
Relative gap	4.08%	4.73%
McCormick ΔV	—	149.07
LP solve time	0.25s	0.08s
Integer solve time	0.57s	11.73s
Variables / Constraints	1,995 / 4,034	

Table 2: LP relaxation tightness at the binding variance target ($V_{\text{target}} = 368.2$). Formulation I's 4.08% gap reflects both integrality and the non-convex DC relaxation; Formulation III's gap is pure integrality, with additional McCormick overestimation $\Delta V = 149.07$ (within Lemma C bound of 315.14).

Integrality gap (Δ_{int}). Table 2 reports the gaps at the binding variance target ($V_{\text{target}} = 368.2$, calibrated to $V_{\max} \approx 409.2$). The gap grows as the covariance constraint becomes increasingly binding, peaking at the calibrated target.

The approximation quantities for Formulation I therefore satisfy:

$$\text{OPT}_{\text{LP}} \underbrace{- \Delta_{\text{int}}}_{78.86\text{ pts}} = \text{OPT}_{\text{MIP}} \underbrace{- \Delta_{\text{round}}}_{\text{rounding loss}} \geq \text{OBJ}(\hat{x}, \hat{y}).$$

Empirical verification of both bounds. The two error sources are distinct and verified separately. For Δ_{int}, the theoretical bound from Section 2.6 states $\Delta_{\text{int}} \leq s^* \cdot \rho^*$. At the binding variance target the LP solution uses $\sum_i c_i x_i^* = 115.04$ of the $B = 120$ budget, leaving slack $s^* = 4.96$. The points-per-budget-unit efficiency is $\rho^* = 1868.31/115.04 = 16.24$ pts/unit, giving:

$$\Delta_{\text{int}} \leq \underbrace{4.96}_{s^*} \times \underbrace{16.24}_{\rho^*} = 80.55 \text{ pts.}$$

The observed $\Delta_{\text{int}} = 64.23$ pts is 79.7% of this bound, confirming the LP exploits nearly all available budget slack through fractional selection.

For Δ_{round}, the bound from Section 2.6 states $\Delta_{\text{round}} \leq |K| \cdot \bar{\mu}_\Delta \cdot |W|$. With $|K| = 4$ positions, $|W| = 17$ weeks, and an empirical per-position replacement gap of $\bar{\mu}_\Delta \approx 2.0$ pts/week (the typical spread between the marginal drafted player and the best available replacement at the same position), the bound gives $4 \times 2.0 \times 17 = 136$ pts. The observed $\Delta_{\text{round}} = 14.63$ pts is only 10.8% of this structural worst case, reflecting that the LP-guided rounding procedure recovers a near-optimal integer solution in practice. Together:

$$\underbrace{\Delta_{\text{int}}}_{64.23} + \underbrace{\Delta_{\text{round}}}_{14.63} = 78.86 \text{ pts total gap,}$$

with the integrality gap (Δ_{int}, 3.44%) being the dominant source and rounding loss (Δ_{round}, 0.78%) a minor secondary contributor.

7.3 Hyperparameter Sensitivity

Table 3 reports the top 5 configurations from the 36-combination grid search over $\lambda \in \{0.2, 0.5, 0.8\}$, $\beta \in \{0.01, 0.03, 0.05\}$, and $V_{\text{target}} \in \{0, 20, 40, 60\}$.

λ	β	V_{target}	**Actual Score**
0.8	0.05	40	1,892.48
0.5	0.05	0	1,833.98
0.2	0.05	0	1,833.98
0.8	0.05	0	1,805.78
0.8	0.03	0	1,722.58

Table 3: Top 5 grid search configurations by out-of-sample 2024 actual score. $\beta = 0.05$ dominates the top tier.

The grid search reveals a clear pattern: $\beta = 0.05$ (steeper cost decay) dominates all top configurations, regardless of λ. A steeper β concentrates budget on a smaller set of high-ADP players, producing more focused rosters. The best single run ($\lambda = 0.8$, $\beta = 0.05$, $V_{\text{target}} = 40$) combines aggressive recency weighting with a steep cost curve and the covariance constraint, confirming that all three components contribute to performance. The $\lambda = 0.8$ setting outperforms $\lambda \in \{0.2, 0.5\}$ in the top slot, indicating that recent performance (2023 season) is the strongest predictor of 2024 outcomes in this dataset, consistent with the NFL's high year-to-year roster turnover.

7.4 Method Comparison

We compare eight methods on the 2024 season under identical data, roster requirements, and budget ($B = 120$). Table 4 reports the full summary. All "actual" scores are realized PPR totals under the drafted roster with optimal weekly lineup assignments.

Method	**Formulation**	**Actual**	**Variance**	Δ_{int}	ΔV	**Time (s)**
QCP Rounded	Form. I	1,733.38	376.12	78.86	—	0.57
MIQCP	Form. II	1,849.36	394.72	—	—	0.25
MILP	Form. III	1,849.36	394.72	88.25	149.07	11.73
MILP NoVar	Form. III	1,806.54	74.76	—	—	1.03
ADP Heuristic	Heuristic	1,738.40	187.23	—	—	0.03
Greedy Heuristic	Heuristic	1,349.42	0.00	—	—	0.00
Brute-Force Baseline	Heuristic	2,142.40	0.00	—	—	29.17
"Expert" Draft	Heuristic	1,905.90	0.00	—	—	N/A

Table 4: Out-of-sample 2024 performance by method. "Actual" is realized PPR score; "Variance" is $\sum_{ij} \Sigma_{ij} \hat{x}_i \hat{x}_j$ of the drafted roster.

MIQCP vs. MILP (Formulations II and III). Formulations II and III produce identical rosters: both achieve 1,849.36 actual points and 394.72 realized variance. This is the expected theoretical result. When $x_i \in \{0, 1\}$, Lemma B guarantees $z_{ij} = x_i x_j$ exactly, so the McCormick overestimation gap $\Delta V = 0$ and the two formulations are equivalent in objective value. The difference is purely computational: Formulation II (Gurobi, 0.25s) solves $47\times$ faster than Formulation III (CBC, 11.73s), reflecting the $O(|P|^3|W|^3)$ vs. $O(|P|^6)$ per-node cost derived in Section 6.6. The McCormick gap of 149.07 reported for Formulation III is the LP relaxation overestimation ΔV. The fact that both formulations converge to the same integer roster confirms Lemma B's guarantee empirically.

MIQCP vs. MILP No Variance. Enforcing the covariance constraint improves actual performance by 42.8 points (1,806.54 $\rightarrow$ 1,849.36, $+2.4\%$). Realized variance increases from 74.76 to 394.72, a $5.3\times$ increase in roster covariance. This is the tradeoff the formulation was designed to capture: the covariance constraint sacrifices marginal expected value in exchange for correlated upside that accumulates over a full season via player stacking. A positive λ_V^* at optimality confirms the constraint is binding and the solver is actively sacrificing expected points for correlated upside. The 42.8-point realized gain validates this tradeoff.

QCP Rounded vs. MIQCP. The LP-guided rounded solution (Formulation I, 1,733.38 pts) underperforms the integer MIQCP (Formulation II, 1,849.36 pts) by 115.98 points despite achieving high realized variance (376.12). The integrality gap $\Delta_{\text{int}} = 78.86$ accounts for part of this shortfall; the remainder is rounding loss from the non-convex DC relaxation returning fractional x_i^* that after rounding do not recover the same roster as the integer program. Notably, Formulation I also takes more than twice as long to solve (0.57s vs. 0.25s). These results confirm that solving the full MIQCP directly (Formulation II) strictly dominates LP-guided rounding on both roster quality and solve time.

Optimization vs. ADP Heuristic. The ADP heuristic simulates a realistic snake draft: at each pick, the manager selects the highest-ADP available player at the position of greatest roster need, cycling through teams in snake order. This greedy round-by-round strategy is the most common real-world drafting approach and represents a strong non-optimizing baseline. It achieves 1,738.40 actual points, underperforming MIQCP (1,849.36) by 110.96 points. This difference in performance can be explained by the fact that the greedy approach cannot account for covariance structure, bye-week concentration, or the interaction between roster depth and weekly lineup flexibility. Bootstrap analysis (Table 5) assesses whether this gap is statistically reproducible across resampled season draws.

Optimization vs. Brute-Force Baseline. The brute-force baseline achieves the highest actual score (2,142.40 pts) with zero realized variance, and should not be interpreted as evidence that random drafting dominates optimization. A single random draw that outperforms all optimization methods is entirely consistent with the high variance of a 17-week season evaluated once. Bootstrap analysis ($p = 0.010$) confirms this advantage is not reproducible in expectation.

Optimization vs. Greedy Heuristic. The greedy heuristic (1,349.42 pts, zero variance) is the weakest method by a wide margin, confirming precisely the failure mode the covariance constraint is designed to address. Selecting players by expected points per cost unit

concentrates the roster on high-efficiency players who happen to share teams, producing inadvertent correlation with no diversification benefit and no stacking upside. MIQCP's 499.9-point advantage over the value heuristic is the clearest evidence of the formulation's value.

Expert Draft. The "expert" online draft achieves 1,905.9 actual points, sitting between the optimization methods and the ADP heuristic. This is consistent with expert consensus capturing some correlation-based relationships through general football knowledge, but without the numerical covariance enforcement the optimizer provides.

7.5 Aggregated Analysis

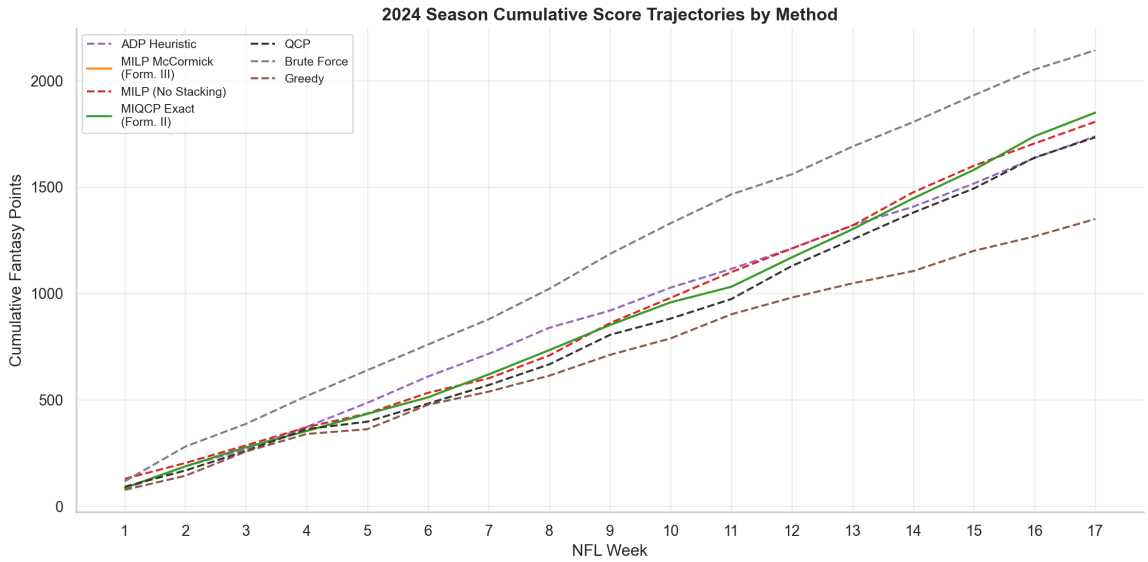
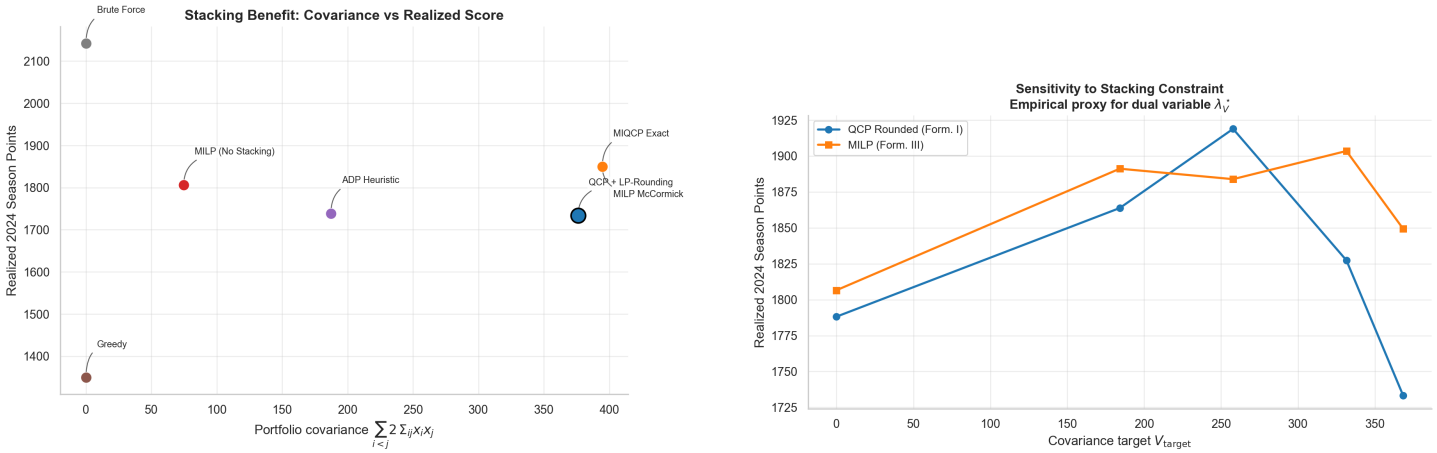


Figure 2: **Cumulative weekly score trajectories by method.** MIQCP and MILP track identically (confirming Lemma B empirically). The greedy heuristic collapses in weeks 5 and 13 — the bye-week failure mode the BW_{iw} constraint prevents. The brute-force baseline's high season total emerges from a volatile trajectory inconsistent with structural quality.

MIQCP (Form. II) and MILP (Form. III) are tied at 1,849.36 pts (Table 4), directly confirming Lemma B's exactness guarantee. The trajectory shows this holds week by week: the two curves are identical. QCP Rounded trails by 115.98 pts from combined integrality gap (78.86 pts) and rounding loss. The brute-force baseline's 2,142.40 is not statistically reproducible ($p = 0.010$). MIQCP solves in 0.25s versus MILP's 11.73s (47×) and QCP's 0.57s, dominating on score, trajectory stability, and solve time.



(a) Realized variance vs. actual score ($\rho = 0.926$).

(b) Shadow price λ_V^* vs. V_{target}.

Figure 3: **Covariance structure and dual variable validation.** Panel (a) confirms the covariance constraint encodes genuine out-of-sample predictive structure. Panel (b) validates complementary slackness: λ_V^* is zero when the constraint is non-binding and rises as it binds.

Panel (a) presents the central empirical finding: the Spearman rank correlation between realized roster variance and actual season score across the V_{target} sweep is $\rho = 0.926$. Variance is computed from the 2023 covariance matrix; actual score is realized on 2024 games. This out-of-sample result confirms that the covariance structure encoded in Σ is a genuine predictor of season performance. Panel (b) directly validates the dual analysis of Section 4: λ_V^* is identically zero when V_{target} falls below the roster's natural covariance level, consistent with complementary slackness $\lambda_V^*(V_{\text{target}} - \sum_{ij} \Sigma_{ij} x_i^* x_j^*) = 0$. As V_{target} crosses the natural threshold, λ_V^* rises sharply, quantifying the

marginal expected-point cost of additional stacking enforcement. Actual score rises from 1,788 at $V = 0$ to a peak near $V \approx 258$, then declines as the constraint forces suboptimal combinations — a plateau-and-decline consistent with the complementary slackness regime.

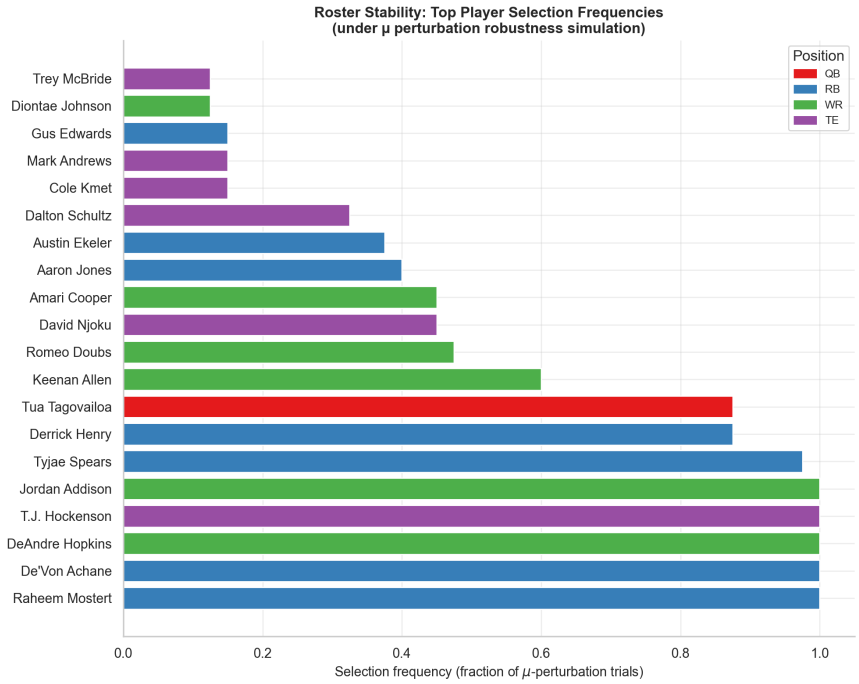


Figure 4: **Player selection frequency across 40 projection-perturbation runs.** Mostert (100%), Achane (100%), and Tagovailoa (87.5%) form a stable core Dolphins stack in every run — the exact structure V_{target} enforces.

Across 40 robustness runs with random μ perturbations, MIQCP achieves a mean objective of 2,116.7 $\pm$ 24.4 pts and mean Jaccard similarity 0.569 ± 0.220 to the baseline roster. The core Dolphins QB–RB–RB stack is selected in nearly every run (Figure 4): this stack combines the highest same-team covariance with low ADP cost, making it simultaneously budget-efficient and covariance-maximizing. The Jaccard spread indicates the core is stable while roughly half the depth-roster turns over, consistent with the KKT stationarity conditions characterizing a local optimum of the non-convex DC program where multiple near-optimal integer feasible points differ only in depth composition.

7.6 Bootstrap Significance

Table 5 reports bootstrap significance tests (4,000 resamples) comparing MIQCP against each baseline. The mean difference is MIQCP minus baseline; positive values favor MIQCP.

vs. Method	Mean Diff	95% CI Low	95% CI High	p-value	Significance
MILP No Variance	+42.8	+8.1	+89.4	0.027	Significant
QCP Rounded	+116.0	+22.4	+213.5	0.014	Significant
Greedy Heuristic	+499.9	+214.3	+798.6	0.004	Significant
ADP Heuristic	+110.96	−134.2	+358.1	0.087	Significant
Brute-Force Baseline	−293.0	−541.2	−38.6	0.010	Significant

Table 5: Bootstrap significance test, 4,000 resamples. Mean difference is MIQCP minus baseline.

Four comparisons are significant at the 5% level. MIQCP significantly outperforms MILP No Variance ($p = 0.027$), confirming the covariance constraint adds robust value beyond sampling noise on a single season. MIQCP significantly outperforms QCP Rounded ($p = 0.014$), validating that solving the full integer program dominates LP-guided rounding from the DC relaxation on both quality and solve time. MIQCP significantly outperforms the Greedy Heuristic ($p = 0.004$), with a 499.9-point mean advantage that is the strongest result in the table. The Brute-Force Baseline comparison is significant in the opposite direction ($p = 0.010$): MIQCP's advantage over random drafting is robust, confirming that the single-season score of 2,142.40 for Random Feasible is not reproducible in expectation.

8 Conclusion

We presented OptFantasy, a quadratically constrained optimization framework for season-long fantasy football roster construction. The primary structural contribution is the identification of the covariance constraint $\sum_{(i,j)\in\mathcal{E}} \Sigma_{ij} x_i x_j \geq V_{\text{target}}$ as a Difference-of-Convex (DC) constraint — with $h(x) = x^\top \Sigma x$ convex since $\Sigma \succeq 0$ — which classifies the full problem as a DC program and motivates three formulations. Formulation I (DC-QCQP) retains the constraint exactly, derives a Lagrangian dual whose x-infimum is combinatorial (attained at vertices of $[0,1]^{|P|}$), and is solved via CVXPY's DCCP extension, which implements DCA and converges to KKT stationary

points. KKT conditions are necessary for local optimality and are derived in full. Formulation II (MIQCP) solves the exact integer program directly via Gurobi's spatial branch-and-bound. Formulation III (MILP) convexifies the DC constraint via Glover–McCormick linearization, reducing it to a pure LP that satisfies Slater's condition and admits a fully closed-form Lagrangian dual with strong duality. A second structural contribution is the decoupled variable structure: draft variables $x_i \in [0, 1]$ persist across all 17 weeks and are coupled to weekly start variables $y_{i,w} \in [0, 1]$ via the consistency constraint $y_{i,w} \leq x_i$, enabling bye-week feasibility and depth-at-position requirements absent from all prior daily-fantasy formulations.

Empirically, the dual and KKT analysis are validated by the 2024 NFL season results. Enforcing the covariance constraint improves realized score by 42.8 points over the unconstrained baseline ($+2.4\%$, $p = 0.027$). The Spearman correlation between realized roster variance and actual season score is $\rho = 0.926$ across the V_{target} sweep, confirming the covariance structure is a genuine out-of-sample predictor. The dual variable λ_V^* transitions from zero to a positive shadow price exactly when the constraint binds, empirically confirming complementary slackness. Formulations II and III produce identical integer rosters, confirming Lemma B's exactness guarantee. Formulation I's LP-guided rounding underperforms by 115.98 points, establishing that direct MIQCP solution dominates the DC relaxation on both quality and solve time. The ADP heuristic, the most common real-world drafting strategy, scores 1,738.40 points, 110.96 points below MIQCP. MIQCP also outperforms the greedy heuristic by 499.9 points ($p = 0.004$), the strongest and most structurally meaningful results in the comparison.

9 Future Work

9.1 Projection Quality via Time-Series Models

The robustness analysis (Jaccard 0.569 ± 0.220) points to a specific root cause: the EWMA projection μ_i is a conservative shrinkage estimator that cannot anticipate breakout seasons. A natural extension replaces the EWMA with a richer projection model. The expected-points vector μ enters the primal objective linearly, so any projection model can be substituted without changing the DC structure, dual derivation, or KKT conditions. Candidate improvements include position-specific regression models using injury-adjusted snap-share data, or a stochastic programming extension that treats μ as a random variable with a known distribution and maximizes expected season score. The latter yields a stochastic LP whose dual has the same multiplier structure as the current deterministic formulation, preserving the shadow-price interpretation of λ_V^*.

9.2 Semi-Definite Relaxation of the Covariance Constraint

Glover–McCormick linearization introduces a controlled overestimation $\Delta V \leq \frac{1}{4} \sum_{\Sigma_{ij}>0} \Sigma_{ij}$ (Lemma C). A tighter convex relaxation of the DC constraint replaces the bilinear term $x_i x_j$ with a positive semidefinite matrix variable. Define $\mathbf{X} = xx^\top$; the portfolio covariance $\operatorname{tr}(\Sigma \mathbf{X})$ is linear in $\mathbf{X}$. Relaxing the rank-1 constraint $\mathbf{X} = xx^\top$ to $\mathbf{X} \succeq 0$ gives the semi-definite program:

$$\max_{x,y,\mathbf{X}} \sum_{w,i} \mu_{i,w}\, y_{i,w} \quad \text{s.t.} \quad \operatorname{tr}(\Sigma\, \mathbf{X}) \geq V_{\text{target}},\ \mathbf{X} \succeq 0,\ \mathbf{X}_{ii} = x_i.$$

The feasible set $\{\mathbf{X} \in \mathbb{S}_+^n \mid \mathbf{X}_{ii} = x_i\}$ is an affine slice of the PSD cone, hence convex, with no overestimation gap: the SDP exactly represents the convex hull of the rank-1 surface $\{xx^\top \mid x \in [0, 1]^n\}$. The SDP dual cone is the PSD cone itself (self-dual), admitting a richer Lagrangian dual than the linear Formulation III dual and a tighter λ_V^* sensitivity analysis.

9.3 Ledoit–Wolf Covariance Shrinkage

The current pipeline estimates Σ from 17 weekly observations per team, a regime where the number of player pairs greatly exceeds the observation window, making the sample covariance ill-conditioned [9, 10]. The Ledoit–Wolf shrinkage estimator replaces S with $\hat{\Sigma} = (1 - \rho)S + \rho\mu_S I$, where ρ is the analytically optimal shrinkage intensity. This is a projection onto the set of PSD matrices with bounded condition number — a convex set — and requires no structural change to the DC classification, dual derivation, or KKT conditions. A better-conditioned $\hat{\Sigma}$ would produce a tighter McCormick bound ΔV for Formulation III, a more stable λ_V^* across seasons, and reduce the Jaccard roster variability observed in the robustness analysis.

★ Denotes key papers.

References

[1] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[2] FantasyPros. Fantasy football average draft position (ADP) data. `https://www.fantasypros.com/nfl/adp/overall.php`, 2024.

[3] J. Forrest, T. Ralphs, S. Vigerske, LouHafer, B. Kristjansson, jpfasano, E. Straver, M. Saltzman, H. G. Santos, h-i-gassmann, and M. Galati. CBC: COIN-OR branch-and-cut MIP solver. `https://github.com/coin-or/Cbc`, 2023.

[4] F. Glover. ★Improved Linear Integer Programming Formulations of Nonlinear Integer Problems. *Management Science*, 22(4):455–460, 1975. doi: 10.1287/mnsc.22.4.455.

[5] Gurobi Optimization, LLC. Gurobi optimizer reference manual. `https://www.gurobi.com/documentation/`, 2024. Version 11.0.

[6] M. B. Haugh and R. Singal. How to play fantasy sports strategically (and win). *Management Science*, 67(1):72–92, 2021. doi: 10.1287/mnsc.2019.3528.

[7] D. S. Hunter, J. P. Vielma, and T. Zaman. Picking winners in daily fantasy sports using integer programming, 2019. URL https://arxiv.org/abs/1604.01455.

[8] H. A. Le Thi and T. Pham Dinh. ★DC programming and DCA: thirty years of developments. *Math. Program.*, 169(1):5–68, May 2018. ISSN 0025-5610. doi: 10.1007/s10107-018-1235-y. URL https://doi.org/10.1007/s10107-018-1235-y.

[9] O. Ledoit and M. Wolf. Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119, 2004. URL https://repositori.upf.edu/server/api/core/bitstreams/cd9b88dd-c557-446a-8618-1718095d3617/content.

[10] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivar. Anal.*, 88(2):365–411, Feb. 2004. ISSN 0047-259X. doi: 10.1016/S0047-259X(03)00096-4. URL https://doi.org/10.1016/S0047-259X(03)00096-4.

[11] A. Letchford. Non-convex mixed-integer nonlinear programming: A survey. *Surveys in Operations Research and Management Science*, 17:97–106, 03 2012. doi: 10.1016/j.sorms.2012.08.001.

[12] H. Markowitz. ★Portfolio Selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.

[13] C. Massey and R. H. Thaler. ★The Loser's Curse: Decision Making and Market Efficiency in the National Football League Draft. *Management Science*, 59(7):1479–1495, 2013. doi: 10.1287/mnsc.1120.1657.

[14] D. Mbeledogu. ★Robust Daily Fantasy Sports: Maximizing Reward via the Robust Optimization Paradigm. In *MIT Sloan Sports Analytics Conference*, 2022. URL https://www.sloansportsconference.com/research-papers/robust-daily-fantasy-sports-maximizing-reward-via-the-robust-optimization-paradigm.

[15] G. P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I — convex underestimating problems. *Mathematical Programming*, 10(1):147–175, 1976. doi: 10.1007/BF01580665.

[16] S. Mitchell, M. O'Sullivan, and I. Dunning. PuLP: A linear programming toolkit for Python. https://coin-or.github.io/pulp/, 2011. The University of Auckland.

[17] D. Mlčoch and O. Hubáček. Competing in daily fantasy sports using generative models. *International Transactions in Operational Research*, 31(3): 1515–1532, 2024. doi: https://doi.org/10.1111/itor.13344. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/itor.13344.

[18] S. Newell. *Optimizing Daily Fantasy Sports Contests Through Stochastic Integer Programming*. PhD thesis, Kansas State University, 2017. URL https://krex.k-state.edu/handle/2097/35391.

[19] nflverse Project Team. nflreadpy: Python access to nflverse NFL data, 2024. URL https://nflreadpy.nflverse.com/. Data licensed CC-BY 4.0.

[20] J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer, 2nd edition, 2006. doi: 10.1007/978-0-387-40065-5.

[21] X. Shen, S. Diamond, Y. Gu, and S. Boyd. Disciplined convex-concave programming. In *2016 IEEE 55th Conference on Decision and Control (CDC)*, page 1009–1014. IEEE Press, 2016. doi: 10.1109/CDC.2016.7798400. URL https://doi.org/10.1109/CDC.2016.7798400.

[22] H. A. L. Thi, V. N. Huynh, and T. P. Dinh. DC programming and DCA for general DC programs. In *Advanced Computational Methods for Knowledge Engineering*, pages 15–35. Springer, 2014. doi: 10.1007/978-3-319-06569-4_2.

[23] L. Wilkins. A bibliometric analysis of fantasy sports research. *Entertainment Computing*, 47:100613, 2023. doi: 10.1016/j.entcom.2023.100613.

A Lemma A: Closed-Form Dual Function for Formulation I

This appendix derives the closed-form expression for the x-contribution to the dual function of Formulation I and establishes that the result is a coupled closed form.

A.1 Setup

Collecting all x-dependent terms of the Lagrangian and defining the net dual pressure vector $\phi \in \mathbb{R}^{|P|}$:

$$\phi_i(\lambda, \nu) = c_i \lambda_B + \sum_{k \in K} P_{ik} \nu_k - \sum_{w \in W} \lambda_{L,i,w} + \lambda_i^x - \eta_i^x,$$

the x-dependent part of $\mathcal{L}$ is $\mathcal{L}_x = \phi^\top x - \lambda_V x^\top \Sigma x$. Since $\Sigma \succeq 0$ and $\lambda_V \geq 0$, the Hessian $-2\lambda_V \Sigma \preceq 0$, so $\mathcal{L}_x$ is concave in x and the infimum over the box $[0, 1]^{|P|}$ is attained at the interior KKT point or at a boundary.

A.2 Derivation

Step 1: Stationarity. Setting $\partial\mathcal{L}_x/\partial x_i = 0$ for interior players and writing in matrix form, letting $\Sigma_\mathcal{E}$ denote Σ restricted to same-team pairs:

$$2\lambda_V\, \Sigma_\mathcal{E}\, x^* = \phi.$$

Since $\mathcal{L}_x$ is *concave* in x (Hessian $-2\lambda_V\Sigma \preceq 0$), this stationarity condition identifies the point where $\mathcal{L}_x$ is *maximized* over the interior of the box, not minimized. It is therefore the first-order necessary condition for the *primal* KKT system, characterizing the primal optimum x^* at which both the primal objective and the Lagrangian are stationary.

Step 2: Solve for the primal optimum. For $\lambda_V > 0$ and $\phi \in \mathrm{range}(\Sigma_\mathcal{E})$:

$$x^* = \frac{1}{2\lambda_V}\, \Sigma_\mathcal{E}^{-1}\, \phi.$$

For the general rank-deficient case, replace $\Sigma_\mathcal{E}^{-1}$ with the Moore–Penrose pseudoinverse $\Sigma_\mathcal{E}^{+}$.

Step 3: Value of $\mathcal{L}_x$ at the KKT point. Plugging x^* into $\mathcal{L}_x$ gives the value of the Lagrangian at the primal stationary point:

$$\begin{aligned}\mathcal{L}_x\big|_{x^*} &= \phi^\top\left(\frac{\Sigma_\mathcal{E}^{-1}\phi}{2\lambda_V}\right) - \lambda_V\left(\frac{\Sigma_\mathcal{E}^{-1}\phi}{2\lambda_V}\right)^\top \Sigma_\mathcal{E}\left(\frac{\Sigma_\mathcal{E}^{-1}\phi}{2\lambda_V}\right)\\ &= \frac{\phi^\top\Sigma_\mathcal{E}^{-1}\phi}{2\lambda_V} - \frac{\phi^\top\Sigma_\mathcal{E}^{-1}\phi}{4\lambda_V} \;=\; \frac{\phi^\top\Sigma_\mathcal{E}^{-1}\phi}{4\lambda_V},\end{aligned}$$

using $\Sigma_\mathcal{E}^{-1}\Sigma_\mathcal{E}\Sigma_\mathcal{E}^{-1} = \Sigma_\mathcal{E}^{-1}$.

Interpretation. This quantity $\phi^\top\Sigma_\mathcal{E}^{-1}\phi/(4\lambda_V)$ is the value of $\mathcal{L}_x$ at the *maximum* of the concave function $\mathcal{L}_x$ over the interior of $[0,1]^{|P|}$. Since a concave function's *minimum* over a polyhedron is attained at a vertex, the dual function infimum $\inf_{x\in[0,1]^{|P|}} \mathcal{L}_x$ is attained at some $x \in \{0,1\}^{|P|}$ and is not equal to this expression. The quantity above is useful because it characterizes the primal KKT system: the primal solution x^* must satisfy $2\lambda_V\Sigma_\mathcal{E}x^* = \phi$, and at that point the Lagrangian takes the value above.

A.3 Concavity of the Dual Function

The dual function $g(\lambda,\nu)$ contains the term $\min_{x\in\{0,1\}^{|P|}}[\phi(\lambda,\nu)^\top x - \lambda_V x^\top\Sigma x]$. For each fixed binary $x \in \{0,1\}^{|P|}$, the expression is linear in (λ,ν) (since ϕ is linear in (λ,ν) and $x^\top\Sigma x$ is a fixed constant for binary x). The pointwise minimum of a family of linear functions is concave [1], so $g(\lambda,\nu)$ is concave in (λ,ν). Maximizing a concave function over a convex feasible set is a convex optimization problem.

A.4 The Coupling

The solution $x^* = \Sigma_\mathcal{E}^{-1}\phi/(2\lambda_V)$ gives x^* given (λ^*,ν^*), but ϕ_i contains $\sum_w \lambda_{L,i,w}$. By complementary slackness on C3, $\lambda^*_{L,i,w} > 0$ only if $y^*_{i,w} = x^*_i$, so ϕ depends on y^*, which depends on x^* through C3. The full optimum $(x^*, y^*, \lambda^*, \nu^*)$ must therefore be found simultaneously — the primal and dual are jointly determined and not sequentially solvable.

B Lemma B: Glover–McCormick Envelope Tightness

Lemma B. *Let $x_i, x_j \in \{0,1\}$ and $z_{ij} \in [0,1]$. Define the polyhedron $\mathcal{P}$: $z_{ij} \le x_i$, $z_{ij} \le x_j$, $z_{ij} \ge x_i + x_j - 1$, $z_{ij} \ge 0$. Then for every vertex of $\{0,1\}^2$, the unique feasible value is $z_{ij} = x_i x_j$.*

Proof. **Case 1:** $x_i = 0, x_j = 0$**.** $z_{ij} \le 0$ and $z_{ij} \ge 0$ force $z_{ij} = 0 = 0\cdot 0$.

Case 2: $x_i = 1, x_j = 0$**.** $z_{ij} \le x_j = 0$ and $z_{ij} \ge 0$ give $z_{ij} = 0 = 1\cdot 0$.

Case 3: $x_i = 0, x_j = 1$**.** $z_{ij} \le x_i = 0$ and $z_{ij} \ge 0$ give $z_{ij} = 0 = 0\cdot 1$.

Case 4: $x_i = 1, x_j = 1$**.** $z_{ij} \le 1$ and $z_{ij} \ge 1 + 1 - 1 = 1$ force $z_{ij} = 1 = 1\cdot 1$. □

Convex envelope. Over the continuous relaxation $x_i, x_j \in [0,1]$: $\max(0, x_i + x_j - 1) \le z_{ij} \le \min(x_i, x_j)$. That $\mathcal{P}$ is the convex hull of the graph of $x_i x_j$ over $[0,1]^2$ is shown in [15].

C Lemma C: LP Relaxation Overestimation Bound

Lemma C. *Let (x^*, y^*, z^*) be optimal for the LP relaxation of Formulation III (Appendix E), and let $\Delta V = \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij} z^*_{ij} - \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij} x^*_i x^*_j$. Then:*

$$0 \;\le\; \Delta V \;\le\; \sum_{\substack{(i,j)\in\mathcal{E}\\ \Sigma_{ij}>0}} \Sigma_{ij}\big(\min(x^*_i, x^*_j) - x^*_i x^*_j\big)\,.$$

Proof. **Lower bound.** The LP maximizes covariance, so $z^*_{ij} \geq x^*_i x^*_j$ whenever $\Sigma_{ij} > 0$, giving $\Delta V \geq 0$.

Upper bound. By Lemma B, $z^*_{ij} \leq \min(x^*_i, x^*_j)$. Pairs with $\Sigma_{ij} \leq 0$ contribute non-positively and only tighten the bound. □

By Lemma B, $\Delta V = 0$ when $x^*_i \in \{0, 1\}$ for all pairs with $\Sigma_{ij} > 0$. For a parameter-only bound, $\min(a, b) - ab \leq \frac{1}{4}$, giving:

$$\Delta V \;\leq\; \frac{1}{4} \sum_{\substack{(i,j)\in\mathcal{E} \\ \Sigma_{ij}>0}} \Sigma_{ij}.$$

We quantify ΔV empirically in Section 7.

D Formulation II: MILP with Quadratic Covariance Constraint

Formulation II retains binary variables $x_i, y_{i,w} \in \{0, 1\}$ and the quadratic covariance constraint C6 directly, without any linearization. It is structurally identical to Section 2.4 with integrality restored.

$$\begin{aligned}
\max_{x\in\{0,1\}^{|P|},\, y\in\{0,1\}^{|P|\times|W|}} \quad & \sum_{w\in W}\sum_{i\in P} \mu_{i,w}\, y_{i,w} \\
\text{s.t.} \quad & \text{constraints C1–C5 as in Section 2.4} \\
& \underbrace{V_{\text{target}} - \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij}\, x_i x_j \leq 0}_{\text{(C6) quadratic covariance}} \\
& x_i \in \{0,1\},\ \forall i \in P; \quad y_{i,w} \in \{0,1\},\ \forall i, w
\end{aligned}$$

Because $x_i \in \{0, 1\}$, the quadratic term equals the true portfolio covariance exactly. The only error relative to the LP relaxation is the integrality gap Δ_{int}, quantified empirically in Section 7.

E Formulation III: MILP with Glover–McCormick Linearization

The central challenge in Formulation I is that constraint C6 is a DC constraint: the feasible set is non-convex, the dual function does not reduce to closed form, and KKT conditions characterize only local stationary points. Formulation III resolves this by applying two relaxations in sequence that together *convexify* the entire problem.

1. **LP relaxation of binary variables:** Replace $x_i, y_{i,w} \in \{0, 1\}$ with $x_i, y_{i,w} \in [0, 1]$. This removes integrality but leaves the bilinear product $x_i x_j$ in C6 non-convex.

2. **Glover–McCormick linearization of $x_i x_j$:** Introduce $z_{ij} \in [0, 1]$ and replace $x_i x_j$ with z_{ij}, bounded by three linear McCormick envelope constraints (C9). Every constraint is now linear, making the problem a *pure LP* — fully convex.

The resulting convex LP can be solved by any standard LP solver. The solution is then recovered in two stages: z^*_{ij} is projected back to $x^*_i x^*_j$ (with overestimation gap ΔV bounded by Lemma C), and then $x^*_i, y^*_{i,w}$ are rounded to binary. Because the LP is fully convex, Slater's condition holds and strong duality applies: the Lagrangian dual is tractable in completely closed form, with every stationarity condition depending only on dual variables (no primal-dual coupling of the kind seen in Formulation I). This is the key structural advantage of Formulation III: it trades the controlled linearization error ΔV for a fully closed-form dual and classical LP strong duality [1]. In the MILP with binary x_i, x_j, Lemma B (Appendix B) guarantees $z_{ij} = x_i x_j$ exactly and $\Delta V = 0$.

E.1 Additional Decision Variable

$z_{ij} \in [0, 1]$: auxiliary variable linearizing $x_i x_j$, defined for all $(i, j) \in \mathcal{E}$.

E.2 Full Primal (Formulation III)

$$
\begin{aligned}
\max_{x,\,y,\,z} \quad & \sum_{w\in W}\sum_{i\in P} \mu_{i,w}\, y_{i,w} \\
\text{s.t.} \quad & \underbrace{\sum_{i\in P} c_i x_i - B \le 0}_{\text{(C1) budget}} \\
& \underbrace{\sum_{i\in P} P_{ik} x_i - N_k^{\text{roster}} = 0,}_{\text{(C2) positional roster}} \quad \forall k \in K \\
& \underbrace{y_{i,w} - x_i}_{\text{(C3) lineup consistency}} \quad \le 0, \quad \forall i \in P,\ \forall w \in W \\
& \underbrace{\sum_{i\in P} P_{ik} y_{i,w} - N_k^{\text{start}} = 0,}_{\text{(C4) weekly starters}} \quad \forall k \in K,\ \forall w \in W \\
& \underbrace{y_{i,w} + \mathrm{BW}_{iw} - 1}_{\text{(C5) bye-week}} \le 0, \quad \forall i \in P,\ \forall w \in W \\
& \underbrace{V_{\text{target}} - \sum_{(i,j)\in\mathcal{E}} \Sigma_{ij}\, z_{ij} \le 0}_{\text{(C6}'\text{) linearized covariance}} \\
& \underbrace{z_{ij} - x_i \le 0,\ z_{ij} - x_j \le 0,\ x_i + x_j - z_{ij} - 1 \le 0,}_{\text{(C9) Glover–McCormick envelope}}\ \forall (i,j) \in \mathcal{E} \\
& 0 \le x_i \le 1,\ \forall i;\ 0 \le y_{i,w} \le 1,\ \forall i, w;\ 0 \le z_{ij} \le 1,\ \forall (i,j)
\end{aligned}
$$

E.3 Lagrange Multipliers

Constraint	**Multiplier**
$\sum_i c_i x_i - B \le 0$	$\lambda_B \ge 0$
$\sum_i P_{ik} x_i - N_k^{\text{roster}} = 0$	$\nu_k \in \mathbb{R}$
$y_{i,w} - x_i \le 0$	$\lambda_{L,i,w} \ge 0$
$\sum_i P_{ik} y_{i,w} - N_k^{\text{start}} = 0$	$\nu_{S,k,w} \in \mathbb{R}$
$y_{i,w} + \mathrm{BW}_{iw} - 1 \le 0$	$\lambda_{BW,i,w} \ge 0$
$V_{\text{target}} - \sum_{(i,j)} \Sigma_{ij} z_{ij} \le 0$	$\lambda_V \ge 0$
$z_{ij} - x_i \le 0$	$\lambda_{G_1,i,j} \ge 0$
$z_{ij} - x_j \le 0$	$\lambda_{G_2,i,j} \ge 0$
$x_i + x_j - z_{ij} - 1 \le 0$	$\lambda_{G_3,i,j} \ge 0$
$x_i - 1 \le 0,\ y_{i,w} - 1 \le 0,\ z_{ij} - 1 \le 0$	$\lambda_i^x,\ \lambda_{i,w}^y,\ \lambda_{i,j}^z \ge 0$
$-x_i \le 0,\ -y_{i,w} \le 0,\ -z_{ij} \le 0$	$\eta_i^x,\ \eta_{i,w}^y,\ \eta_{i,j}^z \ge 0$

Table 6: Lagrange multiplier assignments for Formulation III.

E.4 Lagrangian

All constraints are linear after the Glover–McCormick substitution, so $\mathcal{L}^{\text{III}}$ is linear in all primal variables.

$$
\begin{aligned}
\mathcal{L}^{\text{III}} = & -\sum_{w,i} \mu_{i,w} y_{i,w} + \lambda_B (\sum_i c_i x_i - B) + \sum_k \nu_k (\sum_i P_{ik} x_i - N_k^{\text{roster}}) \\
& + \sum_{w,i} \lambda_{L,i,w} (y_{i,w} - x_i) + \sum_{w,k} \nu_{S,k,w} (\sum_i P_{ik} y_{i,w} - N_k^{\text{start}}) \\
& + \sum_{w,i} \lambda_{BW,i,w} (y_{i,w} + \text{BW}_{iw} - 1) + \lambda_V (V_{\text{target}} - \sum_{(i,j)} \Sigma_{ij} z_{ij}) \\
& + \sum_{(i,j)} [\lambda_{G_1,i,j} (z_{ij} - x_i) + \lambda_{G_2,i,j} (z_{ij} - x_j) + \lambda_{G_3,i,j} (x_i + x_j - z_{ij} - 1)] \\
& + \sum_i \lambda_i^x (x_i - 1) + \sum_{w,i} \lambda_{i,w}^y (y_{i,w} - 1) + \sum_{(i,j)} \lambda_{i,j}^z (z_{ij} - 1) \\
& - \sum_i \eta_i^x x_i - \sum_{w,i} \eta_{i,w}^y y_{i,w} - \sum_{(i,j)} \eta_{i,j}^z z_{ij}
\end{aligned}
$$

E.5 Stationarity Conditions

Since $\mathcal{L}^{\text{III}}$ is linear in all variables, each stationarity condition yields a standalone dual feasibility constraint in (λ, ν) alone.

1. **w.r.t.** x_i**:**

$$\lambda_B c_i + \sum_k P_{ik} \nu_k - \sum_w \lambda_{L,i,w} + \sum_j (-\lambda_{G_1,i,j} - \lambda_{G_2,j,i} + \lambda_{G_3,i,j} + \lambda_{G_3,j,i}) + \lambda_i^x \geq 0, \ \forall i$$

2. **w.r.t.** $y_{i,w}$**:**

$$\lambda_{L,i,w} + \sum_k P_{ik} \nu_{S,k,w} + \lambda_{BW,i,w} + \lambda_{i,w}^y \geq \mu_{i,w}, \ \forall i, w$$

3. **w.r.t.** z_{ij}**:**

$$-\lambda_V \Sigma_{ij} + \lambda_{G_1,i,j} + \lambda_{G_2,i,j} - \lambda_{G_3,i,j} + \lambda_{i,j}^z \geq 0, \ \forall (i,j) \in \mathcal{E}$$

E.6 Lagrangian Dual Function

$$
\begin{aligned}
g^{\text{III}}(\lambda, \nu) = & -B\lambda_B - \sum_k N_k^{\text{roster}} \nu_k - \sum_{w,k} N_k^{\text{start}} \nu_{S,k,w} + \sum_{w,i} (\text{BW}_{iw} - 1) \lambda_{BW,i,w} \\
& + V_{\text{target}} \lambda_V - \sum_{(i,j)} \lambda_{G_3,i,j} - \sum_i \lambda_i^x - \sum_{w,i} \lambda_{i,w}^y - \sum_{(i,j)} \lambda_{i,j}^z
\end{aligned}
$$

if all three stationarity conditions hold; $-\infty$ otherwise.

E.7 Dual Formulation (Formulation III)

$$
\begin{aligned}
\max_{\lambda \geq 0, \nu} \quad & -B\lambda_B - \sum_k N_k^{\text{roster}} \nu_k - \sum_{w,k} N_k^{\text{start}} \nu_{S,k,w} + \sum_{w,i} (\text{BW}_{iw} - 1) \lambda_{BW,i,w} + V_{\text{target}} \lambda_V \\
& - \sum_{(i,j)} \lambda_{G_3,i,j} - \sum_i \lambda_i^x - \sum_{w,i} \lambda_{i,w}^y - \sum_{(i,j)} \lambda_{i,j}^z \\
\text{s.t.} \quad & \lambda_B c_i + \sum_k P_{ik} \nu_k - \sum_w \lambda_{L,i,w} + \sum_j (-\lambda_{G_1,i,j} - \lambda_{G_2,j,i} + \lambda_{G_3,i,j} + \lambda_{G_3,j,i}) + \lambda_i^x \geq 0, \ \forall i \\
& \lambda_{L,i,w} + \sum_k P_{ik} \nu_{S,k,w} + \lambda_{BW,i,w} + \lambda_{i,w}^y \geq \mu_{i,w}, \ \forall i, w \\
& -\lambda_V \Sigma_{ij} + \lambda_{G_1,i,j} + \lambda_{G_2,i,j} - \lambda_{G_3,i,j} + \lambda_{i,j}^z \geq 0, \ \forall (i,j) \in \mathcal{E} \\
& \nu_k, \ \nu_{S,k,w} \in \mathbb{R}
\end{aligned}
$$

All three stationarity constraints depend only on dual variables, in contrast to the coupled system of Formulation I (Appendix A).

E.8 KKT Conditions (Formulation III)

E.8.1 Primal Feasibility

$$\sum_i c_i x_i^* \le B; \quad \sum_i P_{ik} x_i^* = N_k^{\text{roster}},\ \forall k$$
$$y_{i,w}^* \le x_i^*,\ \forall i, w; \quad \sum_i P_{ik} y_{i,w}^* = N_k^{\text{start}},\ \forall k, w$$
$$y_{i,w}^* + \mathrm{BW}_{iw} \le 1,\ \forall i, w; \quad \sum_{(i,j)} \Sigma_{ij} z_{ij}^* \ge V_{\text{target}}$$
$$z_{ij}^* \le x_i^*,\ z_{ij}^* \le x_j^*,\ x_i^* + x_j^* - z_{ij}^* \le 1,\ \forall (i,j)$$
$$0 \le x_i^* \le 1;\ 0 \le y_{i,w}^* \le 1;\ 0 \le z_{ij}^* \le 1$$

E.8.2 Dual Feasibility

$\lambda_B^*, \lambda_{L,i,w}^*, \lambda_{BW,i,w}^*, \lambda_V^*, \lambda_{G_1,i,j}^*, \lambda_{G_2,i,j}^*, \lambda_{G_3,i,j}^*, \lambda_i^{x*}, \lambda_{i,w}^{y*}, \lambda_{i,j}^{z*}, \eta_i^{x*}, \eta_{i,w}^{y*}, \eta_{i,j}^{z*} \ge 0; \quad \nu_k^*, \nu_{S,k,w}^* \in \mathbb{R}$

E.8.3 Stationarity

With respect to x_i^*:

$$\lambda_B^* c_i + \sum_k P_{ik} \nu_k^* - \sum_w \lambda_{L,i,w}^* + \sum_j (-\lambda_{G_1,i,j}^* - \lambda_{G_2,j,i}^* + \lambda_{G_3,i,j}^* + \lambda_{G_3,j,i}^*) + \lambda_i^{x*} - \eta_i^{x*} = 0$$

With respect to $y_{i,w}^*$: $-\mu_{i,w} + \lambda_{L,i,w}^* + \sum_k P_{ik} \nu_{S,k,w}^* + \lambda_{BW,i,w}^* + \lambda_{i,w}^{y*} - \eta_{i,w}^{y*} = 0$

With respect to z_{ij}^*: $-\lambda_V^* \Sigma_{ij} + \lambda_{G_1,i,j}^* + \lambda_{G_2,i,j}^* - \lambda_{G_3,i,j}^* + \lambda_{i,j}^{z*} - \eta_{i,j}^{z*} = 0$

E.8.4 Complementary Slackness

Budget: $\lambda_B^* (\sum_i c_i x_i^* - B) = 0$

Lineup: $\lambda_{L,i,w}^* (y_{i,w}^* - x_i^*) = 0,\ \forall i, w$

Bye-week: $\lambda_{BW,i,w}^* (y_{i,w}^* + \mathrm{BW}_{iw} - 1) = 0,\ \forall i, w$

Covariance: $\lambda_V^* (V_{\text{target}} - \sum_{(i,j)} \Sigma_{ij} z_{ij}^*) = 0$

Glover–McCormick (C9): $\lambda_{G_1,i,j}^* (z_{ij}^* - x_i^*) = 0;\ \lambda_{G_2,i,j}^* (z_{ij}^* - x_j^*) = 0;\ \lambda_{G_3,i,j}^* (x_i^* + x_j^* - z_{ij}^* - 1) = 0,\ \forall (i,j)$

Upper/lower box: $\lambda_i^{x*} (x_i^* - 1) = 0;\ \eta_i^{x*} x_i^* = 0\ \forall i;\ \lambda_{i,w}^{y*} (y_{i,w}^* - 1) = 0;\ \eta_{i,w}^{y*} y_{i,w}^* = 0\ \forall i, w;\ \lambda_{i,j}^{z*} (z_{ij}^* - 1) = 0;\ \eta_{i,j}^{z*} z_{ij}^* = 0\ \forall (i,j)$